

On the Uniqueness of Faithful Embeddings of Causal Sets

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Abstract

We prove that the faithful embedding of a causal set into a globally hyperbolic Lorentzian manifold is essentially unique. If a finite causal set C , obtained by Poisson sprinkling at density ρ , admits faithful embeddings into two globally hyperbolic spacetimes (M_1, g_1) and (M_2, g_2) , then with high probability there exists a diffeomorphism $\Phi: M_1^\circ \rightarrow M_2^\circ$ between interior subdomains that is an ε -approximate isometry, with $\varepsilon \rightarrow 0$ as $\rho \rightarrow \infty$ and $M_i^\circ \nearrow M_i$ in the same limit. The diffeomorphism is constructed as a moving Riemannian centre of mass (Karcher mean) of embedded point correspondences. Its invertibility is established by Poisson covariance concentration and a Lorentzian trilateration argument, and its approximate isometry property follows from mesoscopic preservation of longest-chain distances — a combinatorial invariant of the causal set. The proof works uniformly in all dimensions including $d = 4$.

1 Introduction

Causal set theory [3, 14] posits that spacetime is fundamentally a locally finite partial order, with the continuum manifold (M, g) emerging as an approximation valid above the discreteness scale. The CST doctrine “order plus number gives geometry” encodes the idea that the causal order determines the conformal structure (Malament [12]) and the cardinality of intervals determines the volume element. Together, these should determine the full metric.

The *Hauptvermutung* of causal set theory makes this precise: if the same causal set embeds faithfully into two spacetimes, those spacetimes must be approximately isometric. Recently, Müller [13] formulated two mathematically precise versions of the conjecture and showed that the direct “finite-set” formulation, in which the embeddings preserve only the causal order, is false: he constructs explicit causal sets faithfully embeddable into pairs of non-isometric Lorentzian manifolds. Müller’s countable version, by contrast, requires the embedding to be a *dense* subset of the manifold and works in a limit where the causal set has been pre-completed to a continuum object. Neither result addresses the physically relevant CST setting: causal sets in CST are *finite* structures obtained by Poisson sprinkling at finite density [14], in which the causal order alone is famously insufficient (this is essentially the content of Müller’s negative result), but the combination of order *and* volume information (F2), together with the

asymptotic high-density limit, is expected to suffice. This article addresses precisely that case: the density ρ enters as a parameter, the volume condition (F2) encodes the Poisson statistics, and we obtain quantitative approximate uniqueness with explicit error bounds vanishing as $\rho \rightarrow \infty$.

Our proof has three stages:

- (I) **Construction** (Section 3): a smooth diffeomorphism $\Phi: M_1^\circ \rightarrow M_2^\circ$ between interior subdomains is built from the point correspondences via a moving Riemannian centre of mass. The key combinatorial input — that the longest-chain length in C approximates the Lorentzian proper time in both embeddings — is established here, then used to prove that Φ is a local diffeomorphism.
- (II) **Conformal rigidity** (Section 4): the longest-chain distance preservation, combined with the explicit formula for the Karcher-mean derivative, forces Φ to be an approximate conformal isometry via an Alexandrov–Zeeman stability argument.
- (III) **Volume rigidity** (Section 5): the conformal factor is pinned to 1 by the counting condition of the faithful embedding.

2 Definitions and setup

Definition 2.1 (Causal set). A *causal set* $(C, \preceq)$ is a locally finite partially ordered set: for every $x, y \in C$ with $x \preceq y$, the interval $I(x, y) := \{z \in C : x \preceq z \preceq y\}$ is finite.

Definition 2.2 (Globally hyperbolic spacetime). A Lorentzian manifold (M, g) of dimension d is *globally hyperbolic* if it admits a Cauchy surface and every causal diamond $J^+(p) \cap J^-(q)$ is compact.

Definition 2.3 (Faithful embedding). Let C be a finite causal set and (M, g) a globally hyperbolic Lorentzian manifold of dimension d with curvature scale $\lambda := \min(\text{inj}(M, h), |\text{Rm}[g]|^{-1/2})$, where h is the auxiliary Riemannian metric of Construction 2.11. Fix parameters $\rho > 0$ (density), $K_d > 0$ (tolerance constant), and a dimensional constant $c_* \in (0, 1)$ defining the admissible mesoscopic range.

A map $f: C \rightarrow M$ is a $(\rho; c_*, K_d)$ -*faithful embedding* if:

- (F1) **Order-preservation (exact)**: For all $x, y \in C$, $f(x) \in J_g^-(f(y))$ iff $x \preceq y$.
- (F2) **Scale-dependent uniform density**: For every causal diamond $D = J_g^+(p) \cap J_g^-(q) \subset M$ with proper time

$$\tau_g(p, q) \in [\tau_{\min}, c_* \lambda], \quad \tau_{\min} := c_*^{-1} \rho^{-1/d} (\log \rho V_M)^{2/d}, \quad (1)$$

and at h -distance $\geq c_* \lambda$ from ∂M (or $\partial M = \emptyset$), the count satisfies

$$|f(C) \cap D| - \rho \text{Vol}_g(D) \leq \delta_D \cdot \rho \text{Vol}_g(D), \quad (2)$$

where the tolerance scales with diamond volume as

$$\delta_D := K_d \sqrt{\frac{\log(\rho V_M)}{\rho \text{Vol}_g(D)}}. \quad (3)$$

(F3) **Scale separation:** $\rho\lambda^d \geq c_*^{-2d}(\log \rho V_M)^2$ (so the admissible range is nonempty).

Remark 2.4 (Why scale-dependent tolerance). A Poisson process at intensity $\rho \cdot dV_g$ has count fluctuations $\sqrt{\mu_D}$ at expected count $\mu_D = \rho \text{Vol}_g(D)$, giving relative error $1/\sqrt{\mu_D} = 1/\sqrt{\rho \text{Vol}_g(D)}$. The tolerance δ_D in (3) matches this precisely (up to the $\sqrt{\log \rho V_M}$ factor needed for a uniform union bound over all admissible diamonds). Thus *scale-dependent* δ_D is the natural Poisson tolerance: larger diamonds have tighter relative bounds, exactly as Poisson statistics dictate.

A definition with a single global δ would force the worst-case (smallest diamond) tolerance onto the entire manifold, slowing the convergence rate of the main theorem from polynomial to logarithmic. The scale-dependent form avoids this and preserves the polynomial rate.

For convenience, when invoking F2 at a fixed scale ℓ_* in the proof, we will write $\delta_{\ell_*} = K_d \sqrt{\log(\rho V_M)/(\rho \ell_*^d)}$, the tolerance specialised to that scale.

Remark 2.5 (Interpretation and use). The boundary condition $d_h(D, \partial M) \geq c_* \lambda$ excludes diamonds touching ∂M , where the embedded count would be truncated by the boundary. For manifolds without boundary (spatially compact spacetimes with Cauchy slices, or precompact subdomains of larger spacetimes), this condition is vacuous.

The range (1) is bounded below by the discreteness scale (modulo the log factor needed for Chernoff concentration) so that diamonds contain enough points for Poisson concentration, and bounded above by the curvature scale so that the local-flatness approximation $g = \eta + O((c_* \lambda)^2/\lambda^2)$ is useful and diamonds fit inside strongly-convex neighbourhoods. In the proof, F2 is invoked at two scales: the smoothing scale ℓ in Lemma 3.16 (target covariance), and the optimised scale $\ell_D = \rho^{-1/(d+2)} \lambda^{2/(d+2)}$ in Proposition 5.1 (volume rigidity). Both scales lie inside the admissible range for sufficiently large ρ .

Lemma 2.6 (Poisson sprinklings are faithful embeddings). *Let (M, g) be a globally hyperbolic Lorentzian manifold of dimension d with finite g -volume V_M and curvature scale λ . Fix $\rho > 0$ and a dimensional constant $c_* \in (0, 1)$ such that $\rho\lambda^d \geq c_*^{-2d}(\log \rho V_M)^2$. Let $\Pi \subset M$ be a Poisson process of intensity $\rho \cdot dV_g$, and let $C := \Pi$ with the partial order induced by J_g^- . Then for any constant $K_d > 0$ sufficiently large, the inclusion $f : C \hookrightarrow M$ is a $(\rho; c_*, K_d)$ -faithful embedding with probability at least*

$$1 - (\rho V_M)^{-K'_d}, \quad (4)$$

where $K'_d > 0$ is a dimensional constant growing with K_d .

Proof. (F1) Order-preservation. By construction: the induced partial order on Π is the restriction of J_g^- , so $x \preceq y \iff f(x) \in J_g^-(f(y))$ holds exactly.

¹The specific form of (3) is one of a family of admissible choices. Replacing δ_D with $\delta'_D = K_d(\log \rho V_M)^{1/2}/(\rho \text{Vol}_g(D))^\beta$ for any $\beta \in (0, 1/2]$ gives a different but logically coherent notion of faithful embedding, with the corresponding convergence rate in Theorem 6.1 scaling as $\rho^{-\beta/(d+2) \cdot (\text{something})}$. The choice $\beta = 1/2$ adopted here matches the natural Poisson fluctuation rate and yields the strongest rate consistent with the BB longest-chain bottleneck. See Remark 7.1 in the Discussion for the broader caveat: convergence rates are hypothesis-dependent and the qualitative result ($\varepsilon \rightarrow 0$ as $\rho \rightarrow \infty$) is more robust than any particular polynomial exponent.

(F2) *Scale-dependent uniform density.* The admissible range is $I := [\tau_{\min}, c_*\lambda]$ with $\tau_{\min} := c_*^{-1}\rho^{-1/d}(\log \rho V_M)^{2/d}$. For each admissible diamond D , the count $N_D := |\Pi \cap D|$ is Poisson with mean $\mu_D := \rho \text{Vol}_g(D)$. By the Chernoff bound, for any $t \in (0, 1)$,

$$\Pr(|N_D - \mu_D| \geq t\mu_D) \leq 2\exp(-ct^2\mu_D).$$

Setting $t = \delta_D$ with $\delta_D^2\mu_D = K_d^2 \log \rho V_M$ makes the exponent independent of D :

$$\delta_D^2\mu_D = K_d^2 \log(\rho V_M) \implies \delta_D = K_d \sqrt{\frac{\log \rho V_M}{\mu_D}} = K_d \sqrt{\frac{\log \rho V_M}{\rho \text{Vol}_g(D)}}.$$

This is exactly the scale-dependent tolerance in (3). The per-diamond failure probability is thus $2\exp(-cK_d^2 \log \rho V_M) = 2(\rho V_M)^{-cK_d^2}$, uniformly over all admissible diamonds.

Union bound. The admissible parameter space has covering number $N_* \leq C(\rho V_M)^2$ on a $\tau_{\min}$ -net. F2 holds at all net points simultaneously with probability

$$\geq 1 - 2N_*(\rho V_M)^{-cK_d^2} \geq 1 - (\rho V_M)^{2-cK_d^2}.$$

Choosing K_d large enough that $cK_d^2 - 2 \geq K'_d$ for any prescribed K'_d gives the stated probability bound.

Continuity between net points. A perturbation of (p, q) by $\tau_{\min}$ changes $\text{Vol}_g(D)$ by at most $O(\tau_{\min} \cdot (c_*\lambda)^{d-1})$, a relative change of $O(\tau_{\min}/\lambda) \ll \delta_D$ for our range. The count N_D changes only by points in the symmetric difference, also Poisson-controlled by the same argument.

(F3) *Scale separation.* Follows directly from the hypothesis $\rho\lambda^d \geq c_*^{-2d}(\log \rho V_M)^2$.

Scaling at relevant scales. At the smoothing scale ℓ used in Lemma 3.16, $\delta_\ell = K_d \sqrt{\log \rho V_M / (\rho \ell^d)}$, which vanishes as a power of ρ . At the volume-rigidity scale $\ell_D = \rho^{-1/(d+2)}\lambda^{2/(d+2)}$ used in Proposition 5.1,

$$\delta_{\ell_D} = K_d \sqrt{\frac{\log \rho V_M}{\rho \ell_D^d}} = K_d \sqrt{\log \rho V_M \cdot \rho^{-(2/(d+2))} \lambda^{-d/(d+2) \cdot 1}} = K_d \rho^{-1/(d+2)} \lambda^{-d/(2(d+2))} \sqrt{\log \rho V_M},$$

which matches the volume-rigidity rate. Thus the scale-dependent δ_D preserves the polynomial convergence rate of the main theorem at the scales where F2 is actually invoked. $\square$

Remark 2.7 (Extension to non-compact spacetimes). The finite-volume hypothesis in Lemma 2.6 is technical, not physical: it makes the union bound over diamonds quantitative. For non-compact globally hyperbolic spacetimes (Minkowski, FLRW, etc.), the theorem is applied to a precompact exhaustion $M = \bigcup_n K_n$ with each K_n of finite volume. Restricting the causal set to $C_n := f^{-1}(K_n)$ gives finite causal sets to which our theorem applies, producing local diffeomorphisms $\Phi_n: K_n^\circ \rightarrow \Phi(K_n^\circ)$. Because the construction is local (the Karcher mean at x depends only on embedded points within $2\alpha\ell$ of x), these local diffeomorphisms are compatible on overlaps and patch to a global diffeomorphism in the limit. Physically, this covers all relevant cases: CST sprinklings on Minkowski diamonds, FLRW patches, and de Sitter regions are all finite-volume restrictions of larger non-compact spacetimes.

Remark 2.8 (Boundary layer vanishes in the limit). The interior subdomain M° (defined in (28)) excludes an h -neighbourhood of ∂M of width $4\alpha\ell$. At the optimal smoothing scale $\ell_* = \rho^{-1/(5d)}\lambda^{4/5}$ (Theorem 6.1), this width is $O(\rho^{-1/(5d)}\lambda^{4/5}) \rightarrow 0$ as $\rho \rightarrow \infty$. Hence

$\text{Vol}_g(M \setminus M^\circ)/\text{Vol}_g(M) \rightarrow 0$, and the diffeomorphism $\Phi: M_1^\circ \rightarrow M_2^\circ$ captures an arbitrarily large fraction of the manifold in the high-density limit. The boundary layer is the unavoidable consequence of being unable to sample beyond ∂M , and matches physical intuition: one cannot reconstruct what one has not sampled.

Definition 2.9 (ε -approximate isometry). A diffeomorphism $\phi: (M_1, g_1) \rightarrow (M_2, g_2)$ is an ε -approximate isometry if $\sup_p \|\phi^* g_2(p) - g_1(p)\|_{g_1} \leq \varepsilon$.

Definition 2.10 (Longest chain and Lorentzian distance). For $x \preceq y$ in C , the *longest chain length* $\ell_C(x, y)$ is the maximum k such that there exist $x = z_0 \prec z_1 \prec \cdots \prec z_k = y$ in C . For $p \in J^-(q)$ in (M, g) , the *Lorentzian distance* (proper time) is $\tau(p, q) := \sup_\gamma \int_\gamma \sqrt{-g(\dot{\gamma}, \dot{\gamma})} dt$, where the supremum is over future-directed causal curves from p to q .

Construction 2.11 (Global auxiliary Riemannian metric). A globally hyperbolic (M, g) admits a smooth Cauchy time function $t: M \rightarrow \mathbb{R}$ (Bernal–Sánchez [2]). Set $T := -\nabla t/|\nabla t|_g$ and define $h := g + 2T^\flat \otimes T^\flat$. Then h is a smooth, globally defined, positive-definite metric with $h = \delta + O(L^2/\lambda^2)$ in h -normal coordinates within any region of diameter $L \ll \lambda$. We write h_1, h_2 for the auxiliary metrics on M_1, M_2 .

Remark 2.12 (Independence of Cauchy surface choice). The metrics h_1 and h_2 are constructed independently from the Cauchy time functions on M_1 and M_2 ; no compatibility between the two foliations is required. The Karcher mean (6) uses h_1 only for the source weights $w_k(x) = \exp(-d_{h_1}^2(x, p_k)/\ell^2)$ and h_2 only for the target objective $\sum w_k d_{h_2}^2(y, q_k)$. Within any mesoscopic region, both metrics satisfy $h_i = \delta + O(L^2/\lambda^2)$ in normal coordinates, regardless of the choice of T . The final conclusion $\Phi^* g_2 \approx g_1$ is a statement about the *Lorentzian* metrics g_1, g_2 , which are intrinsic to the spacetimes and independent of any auxiliary choice.

Remark 2.13 (Adapted normal coordinates). For computations involving the Lorentzian structure, we use h_i -normal coordinates with the orthonormal basis chosen so that $e_0 = T_i$. In such adapted coordinates, $h_i = \delta + O(|p|^2/\lambda^2)$ (since they are h_i -normal) and $g_i = \eta + O(|p|^2/\lambda^2)$ (since $T_i^\flat = -dx^0$ in this basis, so $g_i(0) = h_i(0) - 2T_i T_i^T = \delta - 2e_0 e_0^T = \eta$). This dual property — a single coordinate chart in which both the auxiliary Riemannian metric is Euclidean and the Lorentzian metric is Minkowski to leading order — is what allows us to apply both Riemannian Karcher-mean estimates and Lorentzian Alexandrov–Zeeman arguments at the same basepoint.

3 Construction of the diffeomorphism

3.1 The smoothing scale and scale hierarchy

The proof juggles several scales, each playing a distinct role. We summarise them here for quick reference; the table should be consulted as a glossary while reading later proofs.

Definition 3.1 (Scale glossary). The proof uses the following scales, related as $\rho^{-1/d} \ll \ell \ll \alpha \ell \ll \lambda$ (with the specific relationships noted):

Symbol	Meaning	Constraint
$\rho^{-1/d}$	Discreteness scale (mean point spacing)	Smallest meaningful scale
ℓ	Karcher mean smoothing scale	$\rho^{-1/d} \ll \ell \ll \lambda$
α	Cutoff width parameter	$\alpha \geq 16$, dimensionless
$\alpha\ell$	Cutoff support radius	Effective support of $w_k(x)$
ℓ_D	Volume-rigidity scale	$\rho^{-1/(d+2)}\lambda^{2/(d+2)}$, optimised
λ	Curvature scale	$\min(\text{inj}(M, h), \text{Rm}[g] ^{-1/2})$
c_*	F2 admissible range constant	Dimensional, $c_* \in (0, 1)$
V_M	Manifold g -volume	Finite (or precompact)
δ_D	F2 tolerance at scale D	$K_d \sqrt{\log \rho V_M / (\rho V_D)}$
$\delta_\ell, \delta_{\ell_D}$	At smoothing, vol-rig scales	Specialisations of δ_D
ε_τ	Distance preservation error	$O((\rho V_M)^{-1/(2d)} \log^{3/2})$
ε	Final isometry error	Equation (37)

Where each scale appears: The smoothing scale ℓ controls the Karcher mean bandwidth (§3); the cutoff support $\alpha\ell$ localises the Karcher minimisation to a strongly-convex ball (Remark 3.4); the anchor scaffold in the trilateration argument (§3.4) uses anchor positions at distance $\alpha\ell/2$ to $2\alpha\ell$ from the centre; the volume-rigidity scale ℓ_D enters Proposition 5.1 via an optimisation balancing curvature against Poisson statistics.

In Theorem 6.1, ℓ is optimised to $\ell_* = \rho^{-1/(5d)}\lambda^{4/5}$ (Remark 6.4), achieving $\varepsilon \rightarrow 0$ as $\rho \rightarrow \infty$.

3.2 The moving Karcher mean

Let $p_k := f_1(c_k) \in M_1$ and $q_k := f_2(c_k) \in M_2$ denote the embedded points.

Construction 3.2 (The diffeomorphism Φ). Fix a smooth, non-negative, non-increasing bump function $\chi: \mathbb{R} \rightarrow [0, 1]$ with $\chi(s) = 1$ for $s \leq 1$ and $\chi(s) = 0$ for $s \geq 2$. Fix a *cutoff multiplier* $\alpha \geq 16$ with $2\alpha\ell \ll \text{inj}(M_i, h_i)$ for $i = 1, 2$ (which holds since $\ell \ll \lambda$ and $\text{inj} \geq c\lambda$ by Remark 3.4). Define *cutoff Gaussian weights*

$$w_k(x) := \exp(-d_{h_1}^2(x, p_k)/\ell^2) \chi(d_{h_1}(x, p_k)/(\alpha\ell)), \quad (5)$$

and the *moving Karcher mean*

$$\Phi(x) := \arg \min_{y \in M_2} \sum_k w_k(x) d_{h_2}^2(y, q_k). \quad (6)$$

Each weight $w_k(x)$ is supported in the ball $B_{2\alpha\ell}^{h_1}(p_k)$, equals $\exp(-d_{h_1}^2(x, p_k)/\ell^2)$ on $B_{\alpha\ell}^{h_1}(p_k)$, and is smoothly tapered to zero in the annulus $\alpha\ell \leq d_{h_1}(x, p_k) \leq 2\alpha\ell$. Because $2\alpha\ell$ lies strictly inside the injectivity radius of h_1 , the distance function $d_{h_1}(x, p_k)$ is C^∞ on the support of w_k , hence $w_k \in C^\infty(M_1)$.

The cutoff truncation introduces only superexponentially small modifications relative to the pure Gaussian weight: the difference between $\exp(-d_{h_1}^2/\ell^2)$ and w_k is bounded by $e^{-\alpha^2}$ pointwise, and integrates to a contribution $O(\alpha^{d-2}e^{-\alpha^2})$ relative to the total weight (Appendix A). For $\alpha = 16$ in $d = 4$, this is bounded by $4096 \cdot e^{-256} \approx 10^{-108}$, utterly dominated by all other error terms in the proof.

Remark 3.3 (Why cutoff weights). The cutoff serves two purposes. First, the support of w_k is now compact and strictly inside the injectivity radius, so the Karcher mean is well-defined by standard theorems [9, 10] without “effective-support” heuristics. Second, the trilateration argument in Corollary 3.8 requires the “inner core” of points $\{p_k : |p_k| \leq 2\ell\}$ to dominate the empirical cross-covariance by a fixed fraction $\beta \geq 0.9$ of the total weight; without the cutoff, the Gaussian tails would extend further and the inner-core fraction would be smaller. The cutoff at $\alpha\ell$ sharply separates the inner core (where trilateration applies) from the boundary layer (where Gaussian weights are exponentially small), making the proof clean.

Remark 3.4 (Curvature of the auxiliary metric). The Lorentzian curvature scale is λ ($|\text{Rm}[g]| \leq 1/\lambda^2$). The auxiliary Riemannian metric $h = g + 2T^\flat \otimes T^\flat$ differs from g by a smooth rank-1 perturbation. The Riemann tensor of h satisfies $|\text{Rm}[h]| = O(|\text{Rm}[g]| + |\nabla T|^2 + |\nabla^2 T|)$. Since $T = -\nabla t / |\nabla t|_g$ for a smooth Cauchy time function, $|\nabla T| = O(1/\lambda)$ and $|\nabla^2 T| = O(1/\lambda^2)$, giving $|\text{Rm}[h]| = O(1/\lambda^2)$. The injectivity radius of (M, h) is also $\geq c\lambda$ for a dimensional constant $c > 0$. Hence balls of radius $\ell \ll \lambda$ in (M, h) are strongly convex.

3.3 Basic properties: proof sketches

Proposition 3.5 (Well-definedness, smoothness, and derivative bounds). *For ρ sufficiently large:*

- (i) **Well-defined:** *The minimiser in (6) exists and is unique.*
- (ii) **Smooth:** $\Phi \in C^\infty(M_1, M_2)$.
- (iii) **Approximate point matching:** $d_{h_2}(\Phi(p_j), q_j) = O(\ell^2/\lambda^2 + (\rho\ell^d)^{-1/2}\ell)$.
- (iv) **Derivative bounds:** $\|D\Phi\|_{C^0} = O(1)$ and $\|D^2\Phi\|_{C^0} = O(1/\ell^2)$.

Proof. Part (i): Well-definedness. With cutoff weights, only finitely many $w_k(x)$ are nonzero at any given x (those p_k with $d_{h_1}(x, p_k) < 2\alpha\ell$). The corresponding target points $\{q_k : w_k(x) > 0\}$ lie in $f_2(C)$, but their h_2 -distances from a candidate minimum y_* are not a priori bounded — we need to argue the minimum is in a specific region.

Restrict to y in the ball $B := B_{4\alpha\ell}^{h_2}(\bar{q})$ around the unweighted average $\bar{q} := \frac{1}{N} \sum_{w_k > 0} q_k$ (which is well-defined since the support is finite). For $y \in B$ and each active q_k , by Corollary 3.7 (applied to embedded points) plus Proposition 3.5(iii), $d_{h_2}(\bar{q}, q_k) \leq C\alpha\ell$ for some absolute C , hence $d_{h_2}(y, q_k) \leq (4\alpha + C\alpha)\ell \leq C'\alpha\ell \ll \text{inj}(M_2, h_2)$. So all active q_k lie within a single strongly convex ball with y (Remark 3.4), where $d_{h_2}^2(y, q_k)$ is strictly convex in y for each fixed q_k . The weighted sum is therefore strictly convex on B , with a unique minimum. Outside B , the objective grows quadratically (since y moves away from all active q_k), so the global minimum is in B . The Karcher mean $\Phi(x)$ is unique.

Part (ii): Smoothness. The minimiser satisfies the first-order condition

$$F(x, y) := \sum_k w_k(x) \exp_y^{-1}(q_k) = 0, \quad (7)$$

where $\exp_y^{-1}(q_k) = -\frac{1}{2}\nabla_y d^2(y, q_k)$. The Jacobian $D_y F$ is the Hessian of the objective, which is strictly positive-definite (Part (i)).

Each w_k is C^∞ globally on M_1 : on the support of w_k (where $d_{h_1}(x, p_k) < 2\alpha\ell < \text{inj}$), the distance function and the bump function are smooth, and outside the support $w_k \equiv 0$. The exponential map $\exp_y^{-1}$ is C^∞ within the injectivity radius of h_2 , which contains all active q_k for y near the minimum. By the implicit function theorem, the solution $y = \Phi(x)$ of $F(x, \Phi(x)) = 0$ is C^∞ in x .

Part (iii): Approximate point matching. Set $x = p_j$ and work in h_2 -normal coordinates centred at q_j , so $q_j = 0$. In these coordinates, $\exp_y^{-1}(q_k) \approx q_k - y + O(|q_k - y|^2/\lambda^2)$ (curvature correction). The first-order condition becomes

$$\sum_k w_k(p_j)(q_k - \Phi(p_j)) + O(\ell^2/\lambda^2 \cdot \sum w_k) = 0,$$

giving $\Phi(p_j) = \sum_k w_k q_k / \sum_k w_k + O(\ell^2/\lambda^2)$. This is the Gaussian-weighted mean of $\{q_k\}$.

Now, $q_k = q_j + \Delta_k = \Delta_k$ where Δ_k is the displacement from q_j . The source points p_k around p_j form a Poisson process with isotropic Gaussian weights. By the isotropy of the Poisson process and the Gaussian: $\mathbb{E}[\sum w_k \Delta_k / \sum w_k] = 0$ (the leading term cancels by symmetry: for every displacement $+v$, the expected contribution from the mirror displacement $-v$ cancels). The residual is:

- curvature: $O(\ell^2/\lambda^2)$ (from the deviation of the volume element and metric from flat),
- Poisson noise: $O(\ell/\sqrt{\rho\ell^d})$ (standard deviation of the weighted mean of $\rho\ell^d$ points with displacements of size $\sim \ell$).

Part (iv): Derivative bounds. Differentiating (7) implicitly:

$$D_x F + D_y F \cdot D\Phi = 0 \implies D\Phi = -[D_y F]^{-1} D_x F.$$

Scale of $D_y F$: The weights $w_k(x)$ depend only on x , not on y , so the y -derivative hits only the inverse exponential map: $D_y F = \sum_k w_k \partial_y [\exp_y^{-1}(q_k)]$, where ∂_y is the derivative with respect to the basepoint y at fixed target q_k . In flat space, $\exp_y^{-1}(q) = q - y$, so $\partial_y [\exp_y^{-1}(q)] = -I$. In curved space, the Jacobi-field expansion of the exponential map gives $\partial_y [\exp_y^{-1}(q_k)] = -I + O(|\tilde{w}'_k|^2/\lambda^2)$, where the curvature correction is bounded by $\|\text{Rm}\| \cdot |\tilde{w}'_k|^2 = O(\ell^2/\lambda^2)$. Summing, $D_y F = -W[I + O(\ell^2/\lambda^2)]$ where $W := \sum_k w_k$. Hence $[D_y F]^{-1} = -(1/W)[I + O(\ell^2/\lambda^2)]$.

Scale of $D_x F$: With the cutoff weights $w_k(x) = e^{-d^2/\ell^2} \chi(d/(\alpha\ell))$ where $d := d_{h_1}(x, p_k)$, the gradient is

$$\nabla_x w_k = -\frac{2}{\ell^2} e^{-d^2/\ell^2} \chi(d/(\alpha\ell)) \nabla_x (d^2/2) + \frac{1}{\alpha\ell} e^{-d^2/\ell^2} \chi'(d/(\alpha\ell)) \nabla_x d.$$

The first term equals $+\frac{2}{\ell^2} \tilde{w}_k w_k$ on the inner support $d \leq \alpha\ell$ (where $\chi = 1$), with $\tilde{w}_k = \exp_x^{-1}(p_k)$. The second term is supported in the annulus $\alpha\ell \leq d \leq 2\alpha\ell$, where $e^{-d^2/\ell^2} \leq e^{-\alpha^2}$, hence is bounded by $O(e^{-\alpha^2}/\ell)$ pointwise. For $\alpha \geq 16$, this correction is $\leq e^{-256}/\ell$, superexponentially suppressed and absorbed into all subsequent error estimates.

Modulo this superexponentially small correction, component-wise $\partial_{x^j} F^i = +\frac{2}{\ell^2} \sum_k w_k \tilde{w}_k^i \tilde{w}_k^j$, so:

$$D_x F = +\frac{2}{\ell^2} \sum_k w_k \tilde{w}'_k \otimes \tilde{w}_k + O(e^{-\alpha^2}).$$

This is $\frac{2W}{\ell^2} \Sigma_{\text{cross}}$ where $\Sigma_{\text{cross}} = \frac{1}{W} \sum_k w_k \tilde{w}'_k \tilde{w}_k^T$ is the weighted cross-covariance of source and target displacements. Since the displacements are $O(\alpha\ell)$, $\|\Sigma_{\text{cross}}\| = O(\ell^2)$. Hence $\|D_x F\| = O(W)$.

Combining: $\|D\Phi\| = \|[D_y F]^{-1}\| \cdot \|D_x F\| = O(1/W) \cdot O(W) = O(1)$.

For the second derivative: differentiating again introduces $D^2 w_k \sim w_k/\ell^2$ (the second derivative of the Gaussian dominates the bump-function contribution, which is $O(1/(\alpha\ell)^2) \sim O(1/\ell^2)$), giving $\|D^2 \Phi\| = O(1/\ell^2)$. $\square$

3.4 Longest chains and distance preservation

The longest chain length $\ell_C(x, y)$ between causally related elements $x \prec y$ is a combinatorial invariant of the abstract causal set C , with no reference to any embedding. We show that, in any faithful embedding, this combinatorial quantity approximates the Lorentzian proper time. Since $\ell_C(x, y)$ is the same for both embeddings, this gives an approximate distance preservation between M_1 and M_2 that depends only on the causal set. This result will be used in two places: to show that the cross-covariance Σ_{cross} is non-degenerate (Corollary 3.8 below), and to show that Φ is approximately conformal (Section 4).

Longest chain approximates proper time.

Proposition 3.6 (Mesoscopic longest chain). *Let $x \prec y$ in C with $f_i(x), f_i(y)$ contained in a mesoscopic diamond of diameter L and Minkowski volume $V \sim L^d$ in (M_i, g_i) , where $i \in \{1, 2\}$. Then with probability $\geq 1 - O(\log(\rho V)/(\rho V))$:*

$$\left| \frac{\ell_C(x, y)}{(m_d \rho)^{1/d}} - \tau_i(f_i(x), f_i(y)) \right| \leq C_d \left(\frac{L^2}{\lambda^2} + \frac{\log^{3/2}(\rho V)}{(\rho V)^{1/(2d)}} \right) \cdot \tau_i, \quad (8)$$

where m_d is the Bollobás–Winkler–Myrheim–Meyer constant, τ_i is the proper time in (M_i, g_i) , and the rigorously-established convergence exponent $1/(2d)$ follows from Theorems 3 and 9 of Bollobás–Brightwell [4].

Proof. The proof reduces the curved-space problem to flat-space combinatorics via a geometric sandwiching argument.

Step 1: Local flat approximation. Choose a base point midway between $p := f_i(x)$ and $q := f_i(y)$ and introduce Riemann normal coordinates. In these coordinates: $g_{\mu\nu} = \eta_{\mu\nu} - \frac{1}{3} R_{\mu\alpha\nu\beta} z^\alpha z^\beta + O(|z|^3/\lambda^3)$. Since the entire diamond has coordinate radius $\leq L$, the metric perturbation $E := g - \eta$ satisfies, for all vectors $v = (v^0, \vec{v})$:

$$|E(v, v)| \leq \epsilon((v^0)^2 + |\vec{v}|^2), \quad \epsilon := C'_d L^2/\lambda^2. \quad (9)$$

Equivalently: $-(1+\epsilon)(v^0)^2 + (1-\epsilon)|\vec{v}|^2 \leq g(v, v) \leq -(1-\epsilon)(v^0)^2 + (1+\epsilon)|\vec{v}|^2$.

Step 2: Sandwiching the causal diamond. Define two auxiliary Minkowski metrics with modified light-cone angles:

$$\eta^+(v, v) := -\frac{1+\epsilon}{1-\epsilon}(v^0)^2 + |\vec{v}|^2, \quad \eta^-(v, v) := -\frac{1-\epsilon}{1+\epsilon}(v^0)^2 + |\vec{v}|^2.$$

If v is g -causal ($g(v, v) \leq 0$), the lower bound gives $(1-\epsilon)|\vec{v}|^2 \leq (1+\epsilon)(v^0)^2$, hence $\eta^+(v, v) = -\frac{1+\epsilon}{1-\epsilon}(v^0)^2 + |\vec{v}|^2 \leq 0$. Conversely, if v is η^- -causal ($\eta^-(v, v) \leq 0$, i.e., $|\vec{v}|^2 \leq \frac{1-\epsilon}{1+\epsilon}(v^0)^2$), then $g(v, v) \leq -(1-\epsilon)(v^0)^2 + (1+\epsilon) \cdot \frac{1-\epsilon}{1+\epsilon}(v^0)^2 = 0$. Therefore:

$$D_{\eta^-}(p, q) \subseteq D_g(p, q) \subseteq D_{\eta^+}(p, q).$$

Since $\eta^\pm$ are related to η by a rescaling of the time coordinate $t' = \sqrt{(1\pm\epsilon)/(1\mp\epsilon)}t = (1 \pm \epsilon + O(\epsilon^2))t$, the proper times satisfy $\tau_\pm = \tau_g(1 \pm O(L^2/\lambda^2))$.

Step 3: Monotonicity of the longest chain. The same Poisson point set $\{p_k\}$ is used for all three metrics; only the causal order changes. The nesting $D^- \subseteq D_g \subseteq D^+$ implies that every chain valid in $(\{p_k\}, \preceq_{\eta^-})$ is also valid in $(\{p_k\}, \preceq_g)$, and every chain valid in $(\{p_k\}, \preceq_g)$ is valid in $(\{p_k\}, \preceq_{\eta^+})$. Hence

$$\ell_{C_{\eta^-}}(x, y) \leq \ell_{C_g}(x, y) \leq \ell_{C_{\eta^+}}(x, y). \quad (10)$$

This is a *deterministic* inequality for each realisation of the Poisson process.

Step 4: Flat-space Bollobás–Brightwell bound. The bounding sets $C_{\eta^\pm}$ are Poisson sprinklings into Minkowski causal diamonds of volume $V_\pm$. By the standard equivalence (in $d = 2$ via light-cone coordinates, generalised in higher dimensions; see [6]), this is statistically equivalent to a Poisson sprinkling of the unit cube $[0, 1]^d$ at density $n = \rho V_\pm$ with the coordinatewise order, the framework of Bollobás–Brightwell [4].

We invoke two results of [4]. Let $H = H_{n,d}$ denote the longest chain length in the Poisson model.

Theorem 9 (rate of mean convergence): Setting $\mathbb{E}H = c_{n,d}n^{1/d}$ with $c_{n,d} \rightarrow m_d$,

$$m_d \geq c_{n,d} \geq m_d - \frac{12K_d \log^{3/2} n}{n^{1/(2d)} \log \log n}, \quad (11)$$

hence $|\mathbb{E}H - m_d n^{1/d}| \leq 12K_d n^{1/(2d)} \log^{3/2} n / \log \log n$.

Theorem 3 (concentration): For n sufficiently large and $2 < \lambda < n^{1/(2d)} / \log \log n$,

$$\Pr\left(|H - \mathbb{E}H| > \frac{\lambda K_d n^{1/(2d)} \log n}{\log \log n}\right) \leq 4\lambda^2 e^{-\lambda^2}. \quad (12)$$

Setting $\lambda = \sqrt{\log n}$ in (12), the deviation is bounded by $K_d n^{1/(2d)} \log^{3/2} n / \log \log n$ with probability $\geq 1 - 4 \log n / n$. Combining with (11) via the triangle inequality:

$$|H - m_d n^{1/d}| \leq C_d n^{1/(2d)} \log^{3/2} n$$

with probability $\geq 1 - O(\log n / n)$. Substituting $n = \rho V_\pm$ and dividing by $(m_d \rho)^{1/d}$:

$$\left| \frac{\ell_{C_{\eta^\pm}}(x, y)}{(m_d \rho)^{1/d}} - \tau_\pm \right| \leq C'_d \frac{\log^{3/2}(\rho V_\pm)}{(\rho V_\pm)^{1/(2d)}} \tau_\pm.$$

Step 5: Synthesis. With probability $\geq 1 - O(\log(\rho V)/(\rho V))$, the BB bounds hold simultaneously for both bounding diamonds. Substituting $\tau_\pm = \tau_g(1 \pm O(L^2/\lambda^2))$ and applying the squeeze (10):

$$\frac{\ell_{C_g}}{(m_d \rho)^{1/d}} \leq \tau_g(1 + O(L^2/\lambda^2)) + C'_g \tau_g \frac{\log^{3/2}(\rho V)}{(\rho V)^{1/(2d)}},$$

and similarly for the lower bound. Consolidating the error terms gives (8). $\square$

Distance preservation. Since the longest chain length $\ell_C(x, y)$ is a property of the abstract causal set C , it is the *same* for both embeddings. Combined with Proposition 3.6:

Corollary 3.7 (Approximate distance preservation). *For $x \prec y$ in C with both $f_1(x), f_1(y)$ and $f_2(x), f_2(y)$ in mesoscopic diamonds of diameter L and Minkowski volume $V \sim L^d$:*

$$|\tau_1(f_1(x), f_1(y)) - \tau_2(f_2(x), f_2(y))| \leq \varepsilon_\tau \cdot \max(\tau_1, \tau_2), \quad (13)$$

where $\varepsilon_\tau = O(L^2/\lambda^2 + (\rho V)^{-1/(2d)} \log^{3/2}(\rho V)) \rightarrow 0$ as $\rho V \rightarrow \infty$.

Proof. Both proper times approximate $\ell_C(x, y)/(m_d \rho)^{1/d}$ with error ε_τ (Proposition 3.6). Triangle inequality. $\square$

Cross-covariance non-degeneracy. We now use the distance preservation (Corollary 3.7) to establish that the cross-covariance matrix $\Sigma_{\text{cross}}(x) := \frac{1}{W} \sum_k w_k(x) \tilde{w}'_k \tilde{w}_k^T$ (introduced in Proposition 3.5(iv) via the x -derivative of the first-order condition) is non-degenerate. This is the key input to the diffeomorphism proof: source and target covariance non-degeneracy (Lemmas 3.15, 3.16) alone do *not* imply cross-covariance non-degeneracy, as a pathological pairing of source and target points could leave both marginals full rank while the cross-covariance has a null space. Distance preservation rules out such pathological pairings.

Corollary 3.8 (Cross-covariance is non-degenerate). *With probability $\geq 1 - \exp(-c_d \sqrt{\rho})$ uniformly over all $x \in M_1^\circ$,*

$$\sigma_{\min}(\Sigma_{\text{cross}}(x)) \geq c_2 \ell^2,$$

for a constant $c_2 > 0$ depending on $d, \delta_\ell, c_*, \alpha$.

Strategy and roadmap. The proof requires upgrading a fact about *individual* pairwise distances (Corollary 3.7: each embedded pair has approximately preserved proper time) into a fact about the *global linear structure* of the correspondence $p \mapsto q$ (that it is approximately a Lorentz transformation plus translation). Three obstacles must be overcome.

First obstacle: spacelike distances are unavailable. Bollobás–Brightwell [4] gives distance preservation only for *causally related* pairs (chain-length asymptotics). For spacelike-separated pairs there is no direct combinatorial estimator. We solve this by *trilateration*: every point p in the inner core will be located via its timelike distances to a fixed scaffold of $d + 1$ “anchor” points $\{a_m\}$, all of which are chronologically related to p . Spacelike distance preservation is bypassed entirely.

Second obstacle: linearity must be inferred from distance preservation. Classically (Alexandrov–Zeeman), any continuous map of $\mathbb{R}^{1,d-1}$ preserving causal relations is conformal-linear, hence Lorentz up to scale. We need a *finite, approximate, quantitative* version: given $d + 1$ points $\{a_m\} \subset \mathbb{R}^{1,d-1}$ in general position and their images $\{b_m\}$ with approximately preserved pairwise Lorentzian distances, there exists a Lorentz transformation $\hat{\Lambda} \in O(1, d - 1)$ such that $b_m \approx \hat{\Lambda} a_m$ pointwise. This is a finite-dimensional Procrustes-type lemma.

Third obstacle: covering all relevant source points. Once the scaffold $\{a_m\} \mapsto \{b_m\}$ is rigidified by a single $\hat{\Lambda}$, every other source point p in the inner core is located by its trilateration coordinates (Lorentzian distances to the anchors). These distances are preserved by Corollary 3.7, so the target q is forced to satisfy $q \approx \hat{\Lambda} p$ pointwise.

Conclusion. With $q \approx \hat{\Lambda}p$ pointwise on the inner core, the empirical cross-covariance reduces: $\Sigma_{\text{cross}}^{\text{core}} = \hat{\Lambda} \Sigma_1^{\text{core}} + (\text{small})$. The source covariance Σ_1^{core} is non-degenerate by isotropy of the Poisson process (Lemma 3.15), and $\hat{\Lambda}$ has bounded singular values (because both source and target covariances are isotropic at the same scale, forcing $\hat{\Lambda} \hat{\Lambda}^T \approx I$). The bound on $\Sigma_{\text{cross}}^{\text{core}}$ transfers to the full Σ_{cross} because the inner core dominates by Gaussian-weight concentration. The proof is therefore organised as follows:

- **Lemma 3.9:** properties of the target anchor configuration (causal structure, well-conditioned basis).
- **Lemma 3.10:** the algebraic identity recovering p from η -distances to the anchors.
- **Lemma 3.11:** with high probability, Poisson points exist near the target anchor configuration.
- **Lemma 3.12:** the finite Lorentzian Procrustes result—approximate distance preservation rigidifies into approximate $\hat{\Lambda}$.
- **Lemma 3.13:** trilateration error analysis—once $\hat{\Lambda}$ is fixed on anchors, pointwise residuals on the inner core are $O(\varepsilon_\tau \alpha \ell)$.
- **Final assembly:** cross-covariance bound via $\Sigma_{\text{cross}}^{\text{core}} = \hat{\Lambda} \Sigma_1^{\text{core}} + (\text{small})$, Weyl’s inequality, and inner-core dominance.

We work throughout in h_1 -normal coordinates at $x = 0$ in M_1 and h_2 -normal coordinates at $\Phi(0) = 0$ in M_2 . In these coordinates, $g_i = \eta + O(\ell^2/\lambda^2)$ and $h_i = \delta + O(\ell^2/\lambda^2)$ on the cutoff support $|p| \leq 2\alpha\ell$.

The anchor configuration. Fix once and for all the following target geometric configuration of $d + 1$ anchor positions in $\mathbb{R}^{1,d-1}$:

$$a_0^* := -\alpha\ell e_0, \quad a_d^* := +\alpha\ell e_0, \quad a_m^* := \alpha\ell e_0 + \frac{\alpha\ell}{4} e_m \quad (m = 1, \dots, d-1), \quad (14)$$

where $\{e_0, e_1, \dots, e_{d-1}\}$ is the standard Minkowski basis (e_0 timelike, e_m spatial). The displacement vectors from a_0^* are

$$v_d^* := a_d^* - a_0^* = 2\alpha\ell e_0, \quad v_m^* := a_m^* - a_0^* = 2\alpha\ell e_0 + \frac{\alpha\ell}{4} e_m \quad (m = 1, \dots, d-1). \quad (15)$$

Lemma 3.9 (Properties of the target configuration). *Assume $\alpha \geq 16$. The configuration (14) satisfies:*

- (C1) **Pairwise causal relations:** a_0^* is in the strict chronological past of every other a_m^* ; specifically, $\eta(a_m^* - a_0^*, a_m^* - a_0^*) \leq -(2\alpha\ell)^2 + (\alpha\ell/4)^2 < 0$.
- (C2) **Inner-core coverage:** Every $p \in \mathbb{R}^d$ with $|p| \leq 2\ell$ satisfies $\eta(a_m^* - p, a_m^* - p) < 0$ for each $m = 0, \dots, d$ (i.e., p is strictly chronologically related to every anchor).
- (C3) **Well-conditioned basis:** The matrix V^* with rows $(v_m^*)^T$ is invertible with $\|V^*\|, \|(V^*)^{-1}\| = \Theta(\alpha\ell)$, hence condition number $\kappa(V^*) = O(1)$.

Proof. (C1): For $m = d$, $\eta(v_d^*, v_d^*) = -(2\alpha\ell)^2$. For $m \in \{1, \dots, d-1\}$, $\eta(v_m^*, v_m^*) = -(2\alpha\ell)^2 + (\alpha\ell/4)^2 = -\alpha^2\ell^2(4 - 1/16) < 0$.

(C2): Fix p with $|p| \leq 2\ell$. For $m = 0$: $a_0^* - p = -\alpha\ell e_0 - p$, so $\eta(a_0^* - p, a_0^* - p) = -(\alpha\ell + p^0)^2 + |\vec{p}|^2$. Since $|p^0|, |\vec{p}| \leq 2\ell$, $(\alpha\ell + p^0)^2 \geq (\alpha\ell - 2\ell)^2 = (\alpha - 2)^2\ell^2$, and $|\vec{p}|^2 \leq 4\ell^2$. For $\alpha \geq 16$, $(\alpha - 2)^2 \geq 196 > 4$, so the displacement is chronological. Symmetrically for $m = d$.

For $m \in \{1, \dots, d-1\}$: $a_m^* - p = (\alpha\ell - p^0)e_0 + (\alpha\ell/4 - p^m)e_m - \sum_{j \neq m} p^j e_j$. The time component squared is $(\alpha\ell - p^0)^2$; the spatial-component squared is $(\alpha\ell/4 - p^m)^2 + \sum_{j \neq m} (p^j)^2 \leq (\alpha\ell/4 + 2\ell)^2 + 4\ell^2$. For $\alpha \geq 16$: $(\alpha\ell - p^0)^2 \geq (\alpha - 2)^2\ell^2$, while $(\alpha\ell/4 + 2\ell)^2 + 4\ell^2 \leq (\alpha/4 + 2)^2\ell^2 + 4\ell^2$. The chronological condition $(\alpha - 2)^2 > (\alpha/4 + 2)^2 + 4$ holds for $\alpha \geq 16$ (LHS ≥ 196 , RHS ≤ 40).

(C3): The matrix V^* has the explicit form

$$V^* = \begin{pmatrix} 2\alpha\ell & \alpha\ell/4 & 0 & \cdots & 0 \\ 2\alpha\ell & 0 & \alpha\ell/4 & \cdots & 0 \\ \vdots & & & \ddots & \\ 2\alpha\ell & 0 & 0 & \cdots & \alpha\ell/4 \\ 2\alpha\ell & 0 & 0 & \cdots & 0 \end{pmatrix}$$

(rows $v_1^*, \dots, v_{d-1}^*, v_d^*$, columns $e_0, e_1, \dots, e_{d-1}$). Subtracting the last row from the first $d-1$ shows the matrix is invertible with explicit form; the condition number is bounded by a dimensional constant (no α dependence after rescaling). $\square$

Lorentzian trilateration identity.

Lemma 3.10 (Trilateration identity). *Let $a_0, \dots, a_d \in \mathbb{R}^d$ with $a_0 = 0$, set $v_m := a_m$ for $m = 1, \dots, d$, and assume $\{v_m\}$ is a basis of $\mathbb{R}^d$. For any $p \in \mathbb{R}^d$, define $D_m(p) := \eta(p - a_m, p - a_m)$. Then the Lorentzian inner products $\eta(p, v_m)$ satisfy*

$$2\eta(p, v_m) = \eta(v_m, v_m) - (D_m(p) - D_0(p)), \quad m = 1, \dots, d. \quad (16)$$

Equivalently, $p = (V\eta)^{-1}\mathbf{w}(p)$ where V has rows v_m^T and $\mathbf{w}(p)_m = \frac{1}{2}[\eta(v_m, v_m) - D_m(p) + D_0(p)]$.

Proof. Expanding $D_m(p) - D_0(p) = \eta(p - a_m, p - a_m) - \eta(p, p) = -2\eta(p, v_m) + \eta(v_m, v_m)$ gives (16). The map $p \mapsto (\eta(p, v_m))_m = V\eta p$ is invertible since V is invertible (basis hypothesis) and η is non-degenerate. $\square$

Existence of the anchor scaffold from Poisson points.

Lemma 3.11 (Anchor existence). *Assume $\alpha \geq 16$. With probability $\geq 1 - (d+1)\exp(-c_d\rho\ell^d)$, there exist embedded points $a_0, \dots, a_d \in \mathcal{P} := f_1(C) \cap \text{supp}(w(x, \cdot))$ satisfying $|a_m - a_m^*|_{h_1} \leq \ell/4$ for each $m = 0, \dots, d$.*

For any such anchors, properties (C1)–(C3) of Lemma 3.9 continue to hold (with slightly relaxed constants), and additionally:

(C4) *Every $p \in B_{2\ell}^{h_1}(0) \cap \mathcal{P}$ is strictly chronologically related (timelike) to every a_m .*

Proof. Existence: The h_1 -balls $B_{\ell/4}(a_m^*)$ are pairwise disjoint (their centres are separated by $\geq \alpha\ell/4 \geq 4\ell \gg \ell/2$) and each has h_1 -volume $\Theta(\ell^d)$. By Poisson statistics, the probability that any given ball is empty is $\exp(-c_d\rho\ell^d)$. Union bound gives the stated probability.

Stability of properties: Perturbations of size $\ell/4$ in each a_m shift the relevant quadratic forms by $O(\alpha\ell \cdot \ell) = O(\alpha\ell^2)$, which is small compared to the gaps in (C1) and (C2) ($\sim (\alpha\ell)^2$). Hence (C1)–(C3) hold with constants modified by $O(1/\alpha)$, still dimensional.

(C4) follows from (C2) applied to the perturbed anchors: the inner-core condition $|p| \leq 2\ell$ together with anchor perturbations $\leq \ell/4$ preserves the chronological-relation gaps of (C2) up to factors of $O(1)$. $\square$

Approximate rigidity from anchor distances.

Lemma 3.12 (Finite Lorentzian Procrustes). *Let $\{a_0, \dots, a_d\} \subset \mathbb{R}^d$ satisfy properties (C1)–(C3) of Lemma 3.9 (we may take $a_0 = 0$ by translation). Let $\{b_0, \dots, b_d\} \subset \mathbb{R}^d$ with $b_0 = 0$. Suppose*

$$|\eta(b_i - b_j, b_i - b_j) - \eta(a_i - a_j, a_i - a_j)| \leq \varepsilon \quad \text{for all } 0 \leq i, j \leq d, \quad (17)$$

with $\varepsilon \leq c_0(\alpha\ell)^2$ for c_0 sufficiently small. Then there exists $\hat{\Lambda} \in O(1, d-1)$ such that

$$\|b_m - \hat{\Lambda}a_m\| \leq C_d \frac{\varepsilon}{\alpha\ell} \quad \text{for each } m = 0, \dots, d, \quad (18)$$

where C_d is a dimensional constant. With $\varepsilon = \varepsilon_\tau(\alpha\ell)^2$ this reads $\|b_m - \hat{\Lambda}a_m\| \leq C_d \varepsilon_\tau \alpha\ell$.

Proof. Identify each $a_m, b_m \in \mathbb{R}^d$ as a column vector, and form the $d \times d$ matrices

$$A := [a_1 \ a_2 \ \dots \ a_d], \quad B := [b_1 \ b_2 \ \dots \ b_d]$$

whose m -th columns are a_m, b_m respectively. By (C3), A is invertible with $\|A^{-1}\| = O(1/(\alpha\ell))$. Define the η -Gram matrices

$$G^a := A^T \eta A, \quad G^b := B^T \eta B,$$

so $G_{ij}^a = a_i^T \eta a_j = \eta(a_i, a_j)$ and analogously for G^b .

Step 1 (Gram matrix preservation). By polarisation, $2\eta(a_i, a_j) = \eta(a_i, a_i) + \eta(a_j, a_j) - \eta(a_i - a_j, a_i - a_j)$ and analogously for b . Applying (17) to the three squared distances yields $|\eta(a_i, a_j) - \eta(b_i, b_j)| \leq \frac{3}{2}\varepsilon$ for all $1 \leq i, j \leq d$, so

$$\|G^a - G^b\| \leq \frac{3d}{2}\varepsilon.$$

Step 2 (defect of the candidate Lorentz transformation). Define the candidate Lorentz map $\Lambda := BA^{-1}$. Then $B = \Lambda A$, equivalently $b_m = \Lambda a_m$ for each m . Compute

$$\Lambda^T \eta \Lambda = (BA^{-1})^T \eta (BA^{-1}) = A^{-T} B^T \eta B A^{-1} = A^{-T} G^b A^{-1}.$$

For an exact Lorentz transformation, $\Lambda^T \eta \Lambda = \eta$; this would require $G^b = A^T \eta A = G^a$. The deviation is

$$\Lambda^T \eta \Lambda - \eta = A^{-T} (G^b - G^a) A^{-1},$$

so

$$\|\Lambda^T \eta \Lambda - \eta\| \leq \|A^{-1}\|^2 \cdot \|G^b - G^a\| = O(\varepsilon/(\alpha\ell)^2). \quad (19)$$

Step 3 (projection to the Lorentz group). We apply Lemma 4.1 to Λ . That lemma's hypothesis is approximate null-cone preservation: $|\eta(\Lambda v, \Lambda v)| \leq \theta' |\Lambda v|^2$ for all null v . Bounding the left side via (19):

$$|\eta(\Lambda v, \Lambda v)| = |v^T (\Lambda^T \eta \Lambda - \eta) v| \leq \|\Lambda^T \eta \Lambda - \eta\| \cdot |v|^2.$$

Bounding $|v|^2$ in terms of $|\Lambda v|^2$: $|v|^2 \leq |\Lambda v|^2 / \sigma_{\min}(\Lambda)^2$. Combining, $\theta' := \|\Lambda^T \eta \Lambda - \eta\| / \sigma_{\min}(\Lambda)^2$.

Both $\sigma_{\min}(\Lambda)$ and $\|\Lambda\|$ are $\Theta(1)$: $\|\Lambda\| = \|BA^{-1}\| \leq \|B\| \cdot \|A^{-1}\| = O(\alpha\ell) \cdot O(1/(\alpha\ell)) = O(1)$, and $\sigma_{\min}(\Lambda) \geq \sigma_{\min}(B)/\|A\| = \Omega(\alpha\ell)/O(\alpha\ell) = \Omega(1)$ (using $\sigma_{\min}(B) \geq c\alpha\ell$ which follows from distance preservation and the fact that the source matrix A has $\sigma_{\min}(A) \geq c\alpha\ell$ from (C3); a perturbation of size E_a in each b_m doesn't change $\sigma_{\min}(B)$ by more than $O(E_a)$).

Hence $\theta' = O(\varepsilon/(\alpha\ell)^2)$ and Lemma 4.1 yields $\hat{\Lambda} \in O(1, d-1)$ with

$$\|\Lambda - \hat{\Lambda}\| \leq C_d \|\Lambda\| \theta' = O(\varepsilon/(\alpha\ell)^2).$$

Step 4 (residuals). For each m :

$$\|b_m - \hat{\Lambda} a_m\| = \|(\Lambda - \hat{\Lambda}) a_m\| \leq \|\Lambda - \hat{\Lambda}\| \cdot \|a_m\| = O(\varepsilon/(\alpha\ell)^2) \cdot O(\alpha\ell) = O(\varepsilon/(\alpha\ell)).$$

Substituting $\varepsilon = \varepsilon_\tau(\alpha\ell)^2$ gives $\|b_m - \hat{\Lambda} a_m\| = O(\varepsilon_\tau \alpha\ell)$. $\square$

Extending to all causally-related points.

Lemma 3.13 (Trilateration extension). *Let $\{a_m\}, \{b_m\}, \hat{\Lambda}$ be as in Lemma 3.12, with $\|b_m - \hat{\Lambda} a_m\| \leq E_a$ for each m . Suppose $p \in \mathbb{R}^d$ is chronologically related to every a_m , and $q \in \mathbb{R}^d$ satisfies*

$$|\eta(q - b_m, q - b_m) - \eta(p - a_m, p - a_m)| \leq E_p \quad \text{for } m = 0, \dots, d. \quad (20)$$

Then

$$\|q - \hat{\Lambda} p\| \leq C_d (E_p/(\alpha\ell) + E_a). \quad (21)$$

Proof. By Lemma 3.10, with $a_0 = 0$ as our choice of origin, p is determined by the η -inner products $\eta(p, v_m)$ via $V\eta p = \mathbf{w}(p)$ where $\mathbf{w}(p)_m = \frac{1}{2}[\eta(v_m, v_m) - D_m(p) + D_0(p)]$.

Apply the same identity in the target space: with $b_0 = 0$ and $v'_m = b_m$, $q = (V'\eta)^{-1} \mathbf{w}'(q)$ where V' has rows $(v'_m)^T$ and analogously for $\mathbf{w}'(q)$.

Compute $\hat{\Lambda} p = \hat{\Lambda}(V\eta)^{-1} \mathbf{w}(p) = (V\eta \hat{\Lambda}^{-1})^{-1} \mathbf{w}(p)$. Since $\hat{\Lambda} \in O(1, d-1)$, $\hat{\Lambda}^T \eta \hat{\Lambda} = \eta$, equivalently $\eta \hat{\Lambda}^{-1} = \hat{\Lambda}^T \eta$. Hence $V\eta \hat{\Lambda}^{-1} = V \hat{\Lambda}^T \eta$, the m -th row of which is $v_m^T \hat{\Lambda}^T \eta = (\hat{\Lambda} v_m)^T \eta$.

By hypothesis, $b_m = \hat{\Lambda} a_m + e_m$ with $\|e_m\| \leq E_a$. Hence $v'_m = b_m = \hat{\Lambda} a_m + e_m = \hat{\Lambda} v_m + e_m$ (using $a_0 = b_0 = 0$ so $v_m = a_m$, $v'_m = b_m$). The m -th row of V' is $(v'_m)^T = (\hat{\Lambda} v_m)^T + e_m^T$, so $V' - V \hat{\Lambda}^T$ has rows e_m^T , giving $\|V' - V \hat{\Lambda}^T\| \leq \sqrt{d} \cdot E_a$. Right-multiplying by η (norm 1): $\|V' \eta - V \hat{\Lambda}^T \eta\| = \|V' \eta - V \eta \hat{\Lambda}^{-1}\| \leq \sqrt{d} E_a$.

For $\mathbf{w}(p) - \mathbf{w}'(q)$: each component is $\frac{1}{2}[\eta(v_m, v_m) - \eta(v'_m, v'_m)] - \frac{1}{2}[D_m(p) - D'_m(q)] + \frac{1}{2}[D_0(p) - D'_0(q)]$. The first bracket is bounded by direct estimation:

$$|\eta(v_m, v_m) - \eta(v'_m, v'_m)| \leq \|\eta\| \cdot (\|v_m\| + \|v'_m\|) \cdot \|v_m - v'_m\| \leq 2\|v_m\| \cdot E_a + O(E_a^2) = O(\alpha\ell \cdot E_a)$$

for $E_a \ll \alpha\ell$. The remaining two brackets are bounded by (20): each is $\leq E_p$. Total: $\|\mathbf{w}(p) - \mathbf{w}'(q)\| \leq O(\alpha\ell \cdot E_a + E_p)$.

Now

$$\begin{aligned} q - \hat{\Lambda}p &= (V'\eta)^{-1}\mathbf{w}'(q) - (V\eta\hat{\Lambda}^{-1})^{-1}\mathbf{w}(p) \\ &= (V'\eta)^{-1}[\mathbf{w}'(q) - \mathbf{w}(p)] + [(V'\eta)^{-1} - (V\eta\hat{\Lambda}^{-1})^{-1}]\mathbf{w}(p). \end{aligned}$$

We bound each term.

First term: $\|(V'\eta)^{-1}\| = O(1/(\alpha\ell))$ (since (C3) holds for V' up to $O(E_a/(\alpha\ell))$ perturbation, still $\Theta(\alpha\ell)$ in the relevant norm). Times $\|\mathbf{w}'(q) - \mathbf{w}(p)\| = O(\alpha\ell \cdot E_a + E_p)$ gives $O(E_a + E_p/(\alpha\ell))$.

Second term: Using $X^{-1} - Y^{-1} = X^{-1}(Y - X)Y^{-1}$: $\|(V'\eta)^{-1} - (V\eta\hat{\Lambda}^{-1})^{-1}\| \leq \|(V'\eta)^{-1}\| \cdot \|V\eta\hat{\Lambda}^{-1} - V'\eta\| \cdot \|(V\eta\hat{\Lambda}^{-1})^{-1}\| = O(1/(\alpha\ell)) \cdot O(E_a) \cdot O(1/(\alpha\ell)) = O(E_a/(\alpha\ell)^2)$. Times $\|\mathbf{w}(p)\| = O((\alpha\ell)^2)$ gives $O(E_a)$.

Summing: $\|q - \hat{\Lambda}p\| \leq O(E_a + E_p/(\alpha\ell))$. $\square$

The cross-covariance bound.

Proof of Corollary 3.8. We assemble the four lemmas.

Step A: Anchors and approximate Lorentz transformation. By Lemma 3.11, with probability $\geq 1 - (d+1)\exp(-c_d\rho\ell^d)$ there exist anchors $\{a_m\} \subset \mathcal{P}$ near the target configuration $\{a_m^*\}$, satisfying (C1)–(C4). Let $b_m := (f_2 \circ f_1^{-1})(a_m)$.

Since the anchors are pairwise causally related at proper times $\sim \alpha\ell$, which lies in the admissible range $[c_*^{-1}\rho^{-1/d}(\log \rho V_M)^{2/d}, c_*\lambda]$ for sufficiently large ρ , Corollary 3.7 gives $|\eta(b_i - b_j, b_i - b_j) - \eta(a_i - a_j, a_i - a_j)| \leq C_1\varepsilon_\tau(\alpha\ell)^2$ for each pair.

By Lemma 3.12 with this $\varepsilon = C_1\varepsilon_\tau(\alpha\ell)^2$, there exists $\hat{\Lambda} \in O(1, d-1)$ with $E_a := \|b_m - \hat{\Lambda}a_m\| \leq C_2\varepsilon_\tau\alpha\ell$ for each m .

Step B: Pointwise residual on the inner core. For each $p \in \mathcal{P}^{\text{core}} := B_{2\ell}^{h_1}(0) \cap \mathcal{P}$, by (C4) p is timelike to every anchor. Corollary 3.7 gives $|\eta(q - b_m, q - b_m) - \eta(p - a_m, p - a_m)| \leq C_3\varepsilon_\tau(\alpha\ell)^2$ where $q := (f_2 \circ f_1^{-1})(p)$.

By Lemma 3.13 with $E_p = C_3\varepsilon_\tau(\alpha\ell)^2$:

$$\|q - \hat{\Lambda}p\| \leq C_4(E_a + E_p/(\alpha\ell)) = O(\varepsilon_\tau\alpha\ell).$$

Hence $q = \hat{\Lambda}p + R(p)$ with $\|R(p)\| \leq C_5\varepsilon_\tau\alpha\ell$.

Step C: Inner-core cross-covariance. Define

$$\Sigma_{\text{cross}}^{\text{core}} := \frac{1}{W^{\text{core}}} \sum_{p_k \in \mathcal{P}^{\text{core}}} w_k q_k p_k^T, \quad W^{\text{core}} := \sum w_k.$$

Substituting $q_k = \hat{\Lambda}p_k + R_k$:

$$\Sigma_{\text{cross}}^{\text{core}} = \hat{\Lambda} \Sigma_1^{\text{core}} + E^{\text{core}}, \quad (22)$$

with $\Sigma_1^{\text{core}} = \frac{1}{W^{\text{core}}} \sum w_k p_k p_k^T$ and operator norm $\|E^{\text{core}}\| \leq \max_k \|R_k\| \cdot \max_k \|p_k\| \leq C_5\varepsilon_\tau\alpha\ell \cdot 2\ell = O(\varepsilon_\tau\alpha\ell^2)$.

Step D: Lower bound on $\sigma_{\min}(\hat{\Lambda}\Sigma_1^{\text{core}})$. By Lemma 3.15 restricted to the inner core (a Poisson sample in $B_{2\ell}^{h_1}(0)$), $\sigma_{\min}(\Sigma_1^{\text{core}}) \geq c_{\text{src}}\ell^2$ with probability $\geq 1 - 2d\exp(-c\rho\ell^d)$.

We claim $\sigma_{\min}(\hat{\Lambda}) \geq c_\Lambda > 0$ for a dimensional constant. By Lemma 3.16 applied to the target inner core, $\sigma_{\min}(\Sigma_2^{\text{core}}) \geq c_{\text{tgt}}\ell^2$, where $\Sigma_2^{\text{core}} = \frac{1}{W^{\text{core}}} \sum w_k q_k q_k^T$. From $q_k = \hat{\Lambda} p_k + R_k$:

$$\Sigma_2^{\text{core}} = \hat{\Lambda} \Sigma_1^{\text{core}} \hat{\Lambda}^T + O(\varepsilon_\tau \alpha \ell^2).$$

Hence $\hat{\Lambda} \Sigma_1^{\text{core}} \hat{\Lambda}^T \succeq (c_{\text{tgt}} - O(\varepsilon_\tau \alpha))\ell^2 I$, which for ρ large is $\geq (c_{\text{tgt}}/2)\ell^2 I$. Taking operator norms and using $\|\hat{\Lambda} \Sigma_1^{\text{core}} \hat{\Lambda}^T\| \leq \|\hat{\Lambda}\|^2 \|\Sigma_1^{\text{core}}\|$, and $\sigma_{\min}(\hat{\Lambda} \Sigma_1^{\text{core}} \hat{\Lambda}^T) = \sigma_{\min}(\hat{\Lambda})^2 \sigma_{\min}(\Sigma_1^{\text{core}})$, we get $\sigma_{\min}(\hat{\Lambda})^2 \geq c_{\text{tgt}}/(2c_{\text{src}} \cdot 2) =: c_\Lambda^2 > 0$. (Here $\sigma_{\min}$ refers to the standard Euclidean operator norm.)

Therefore

$$\sigma_{\min}(\hat{\Lambda} \Sigma_1^{\text{core}}) \geq \sigma_{\min}(\hat{\Lambda}) \cdot \sigma_{\min}(\Sigma_1^{\text{core}}) \geq c_\Lambda \cdot c_{\text{src}} \ell^2 =: c_* \ell^2. \quad (23)$$

Step E: Inner-core conclusion. By Weyl's inequality and (22), (23):

$$\sigma_{\min}(\Sigma_{\text{cross}}^{\text{core}}) \geq c_* \ell^2 - \|E^{\text{core}}\| = c_* \ell^2 - O(\varepsilon_\tau \alpha \ell^2) \geq (c_*/2)\ell^2$$

for ρ large enough that $\varepsilon_\tau \alpha \ll c_*$.

Step F: Inner core dominates the full cross-covariance. The full empirical cross-covariance is $\Sigma_{\text{cross}} = \beta \Sigma_{\text{cross}}^{\text{core}} + (1 - \beta) \Sigma_{\text{cross}}^{\text{bdry}}$, where $\beta = W^{\text{core}}/W$ is the inner-core weight fraction.

For Gaussian weights $w(p) = e^{-|p|^2/\ell^2}$ on the ball of radius 2ℓ (inner core), the weight fraction satisfies $\beta \geq 1 - \Gamma(d/2, 4)/\Gamma(d/2)$ where $\Gamma(s, x)$ is the upper incomplete gamma function. For $d = 4$: $\Gamma(2, 4)/\Gamma(2) = 5e^{-4} \approx 0.092$, so $\beta \geq 0.9$.

The boundary contribution $\Sigma_{\text{cross}}^{\text{bdry}}$ involves only $\|q_k\|, \|p_k\| \leq 2\alpha\ell$, so $\|\Sigma_{\text{cross}}^{\text{bdry}}\| \leq C(\alpha\ell)^2$. The boundary contribution $\Sigma_{\text{cross}}^{\text{bdry}}$ satisfies a sharper bound using the smallness of the boundary weights. For $|p| \geq 2\ell$, the Gaussian factor $w(p) \leq e^{-4} < 0.02$, so the weighted contribution to any matrix entry is bounded by $\|q_k\| \cdot \|p_k\| \cdot w_k \leq (2\alpha\ell)^2 \cdot e^{-4} \leq 0.07 \alpha^2 \ell^2$. Averaging, $\|\Sigma_{\text{cross}}^{\text{bdry}}\| \leq C' \ell^2$ for a constant C' depending on d and α but *not* growing with α^2 (because the exponentially-small weight dominates the polynomial α^2 for any fixed $\alpha \geq 16$).

By Weyl's inequality,

$$\sigma_{\min}(\Sigma_{\text{cross}}) \geq \beta \sigma_{\min}(\Sigma_{\text{cross}}^{\text{core}}) - (1 - \beta) \|\Sigma_{\text{cross}}^{\text{bdry}}\| \geq 0.9 \cdot (c_*/2)\ell^2 - 0.1 \cdot C' \ell^2 \geq c_2 \ell^2$$

for $c_2 > 0$ a dimensional constant (the relevant inequality is $0.45c_* \geq 0.1C'$, which holds because C' is controlled by the exponentially-small weights).

Step G: Uniformity over $x \in M_1^\circ$. Cover M_1° by an ℓ -net $\mathcal{N}$ of size $|\mathcal{N}| \leq C \text{Vol}(M_1)/\ell^d$. At each $x \in \mathcal{N}$, the argument above applies with failure probability $\leq C'(d+1) \exp(-c_d \rho \ell^d)$ (from Lemma 3.11, dominating the matrix concentration error). Union bound gives failure probability $\leq C(\text{Vol}(M_1)/\ell^d) \exp(-c_d \rho \ell^d) \leq \exp(-c'_d \rho \ell^d)$ for $\rho \ell^d \gg \log(\text{Vol}(M_1)/\ell^d)$.

For $x \notin \mathcal{N}$: $\Sigma_{\text{cross}}(x)$ varies Lipschitz-continuously in x with constant $\|D_x \Sigma_{\text{cross}}\| = O(\ell)$ (since $\nabla_x w_k = O(w_k/\ell)$ and the matrix entries are $O(\ell^2)$). A perturbation of x by ℓ changes Σ_{cross} in operator norm by $O(\ell^2)$, which is small compared to the bound $c_2 \ell^2$ by a fixed ratio. Hence the bound transfers from net points to all of M_1° , with c_2 replaced by $c_2/2$ (still positive). $\square$

3.5 Diffeomorphism via cross-covariance

We show $D\Phi$ is everywhere invertible. The argument uses three ingredients: (a) the source covariance Σ_1 is non-degenerate (Lemma 3.15), forced by the isotropy of the Poisson sprinkling; (b) the target covariance Σ_2 is non-degenerate (Lemma 3.16), forced by the faithful

embedding conditions; and (c) the cross-covariance Σ_{cross} is non-degenerate (Corollary 3.8), forced by the longest-chain distance preservation that links the source and target point clouds. The first two are properties of the marginal point distributions; the third is a property of the *pairing* between them, which is the genuine output of the faithful embedding hypothesis.

Definition 3.14 (Weighted covariance matrices). At $x \in M_1$, define displacement vectors $\tilde{w}_k := \exp_x^{-1}(p_k) \in T_x M_1$ and $\tilde{w}'_k := \exp_{\Phi(x)}^{-1}(q_k) \in T_{\Phi(x)} M_2$. The *source and target covariance matrices* are

$$\Sigma_1(x) := \frac{\sum_k w_k(x) \tilde{w}_k \tilde{w}_k^T}{\sum_k w_k(x)}, \quad (24)$$

$$\Sigma_2(x) := \frac{\sum_k w_k(x) \tilde{w}'_k \tilde{w}'_k^T}{\sum_k w_k(x)}. \quad (25)$$

Lemma 3.15 (Source covariance is non-degenerate). *With probability $\geq 1 - \exp(-c_d \rho \ell^d)$ uniformly over all $x \in M_1$:*

$$\frac{\ell^2}{4} I \leq \Sigma_1(x) \leq 2\ell^2 I. \quad (26)$$

Proof. Work in h_1 -normal coordinates centred at x , so $p_k \approx \tilde{w}_k$ and $h_1 \approx \delta$ up to $O(\ell^2/\lambda^2)$ corrections. The source points within distance $\sim \ell$ of x form a Poisson process at density ρ in an approximately Euclidean ball. The key property of the Poisson process used here is *independence*: the location of each sprinkled point is statistically independent of every other point. This is essential for the Matrix Bernstein inequality below (a regular lattice or repulsive point process would have correlated positions, invalidating the bound).

The *population* (continuum) covariance with cutoff weight $w(x) = e^{-|x|^2/\ell^2} \chi(|x|/\alpha\ell)$ is

$$\bar{\Sigma} = \frac{\int_{\mathbb{R}^d} x x^T w(x) \rho d^d x}{\int_{\mathbb{R}^d} w(x) \rho d^d x} = \frac{\ell^2}{2} I + O(e^{-\alpha^2} \ell^2),$$

since on the inner support $|x| \leq \alpha\ell$ the cutoff factor $\chi = 1$, and on the annulus $\alpha\ell \leq |x| \leq 2\alpha\ell$ the Gaussian factor $e^{-|x|^2/\ell^2} \leq e^{-\alpha^2}$ produces a correction bounded by $\alpha^d e^{-\alpha^2} \ell^2$ to the integral. The integral $\int x x^T e^{-|x|^2/\ell^2} \rho d^d x = (\ell^2/2)I \cdot W_0$ where W_0 is the corresponding denominator (without the spatial moment), so the ratio is $(\ell^2/2)I$ to leading order, with curvature correction $O(\ell^4/\lambda^2)$ from the volume element. For $\alpha \geq 4$ in $d = 4$, the cutoff correction $O(e^{-16} \ell^2) \approx 1.1 \times 10^{-7} \ell^2$ is negligible compared to all other error terms.

The *empirical* covariance Σ_1 is a sum of independent random matrices $M_k := w_k \tilde{w}_k \tilde{w}_k^T / W$. The centred matrices $Z_k := M_k - \mathbb{E}[M_k]$ are zero-mean but not positive semi-definite, so the appropriate concentration tool is the *Matrix Bernstein inequality* (Tropp [15]): for independent zero-mean random matrices Z_k with $\|Z_k\| \leq B$ and matrix variance $\sigma^2 := \|\sum \mathbb{E}[Z_k Z_k^T]\|$,

$$\Pr[\|\sum Z_k\| \geq t] \leq 2d \exp\left(\frac{-t^2/2}{\sigma^2 + Bt/3}\right).$$

In our setting, $B = O(\ell^2/N_{\text{eff}})$ (each term is bounded by ℓ^2 divided by the number of terms), $\sigma^2 = O(\ell^4/N_{\text{eff}})$, and $t = \ell^2/4$. This gives $\|\Sigma_1 - \bar{\Sigma}\| \leq \frac{1}{4}\ell^2$ with probability $\geq 1 - 2d \exp(-c\rho\ell^d)$. By Weyl's inequality, combined with $\bar{\Sigma} = (\ell^2/2)I$: $\sigma_{\min}(\Sigma_1) \geq \ell^2/2 - \ell^2/4 = \ell^2/4$. A union bound over $O(\text{Vol}(M_1)/\ell^d)$ locations is absorbed by the exponential tail. $\square$

Lemma 3.16 (Target covariance is non-degenerate). *With probability $\geq 1 - \exp(-c_d \rho \ell^d)$ uniformly over all $x \in M_1$:*

$$\sigma_{\min}(\Sigma_2(x)) \geq c_1 \ell^2 \quad (27)$$

for a constant $c_1 > 0$ depending only on d and δ_ℓ , where $\delta_\ell = K_d \sqrt{\log \rho V_M / (\rho \ell^d)}$ is the F2 tolerance at the smoothing scale.

Proof. The target points $\{q_k\}$ with significant weight are the images under f_2 of the causal set elements c_k whose source images $p_k = f_1(c_k)$ are within distance $\sim \ell$ of x . There are $N \sim \rho \ell^d$ such points, with target displacements $\tilde{w}'_k = \exp_{\Phi(x)}^{-1}(q_k) \in T_{\Phi(x)} M_2$.

By the Rayleigh quotient,

$$\sigma_{\min}(\Sigma_2) = \min_{|v|=1} v^T \Sigma_2 v = \min_{|v|=1} \frac{\sum_k w_k (v \cdot \tilde{w}'_k)^2}{\sum_k w_k}.$$

If $\sigma_{\min}(\Sigma_2) < \varepsilon^2$, then there exists a unit vector v such that the weighted average of $(v \cdot \tilde{w}'_k)^2$ is less than ε^2 . We must derive a contradiction with the faithful embedding conditions for any v in $T_{\Phi(x)} M_2$. Two cases arise depending on whether v is timelike or spacelike (the null case is a measure-zero boundary handled by continuity).

Timelike case. Suppose v is a future-directed unit timelike vector at $\Phi(x)$. With probability $\geq 1 - \exp(-c \rho \ell^d)$, the Poisson sprinkling produces chains of length $m \sim (\rho \ell^d)^{1/d}$ among the N source elements: there exist $c_{i_1} \prec c_{i_2} \prec \dots \prec c_{i_m}$ in C with all $f_1(c_{i_j})$ within distance $\sim \ell$ of x . (This is a standard result about chain length in Poisson sprinklings of Minkowski diamonds, used as input to the Brightwell–Gregory theorem.) By (F1), the targets $q_{i_1} \prec \dots \prec q_{i_m}$ are causally ordered in (M_2, g_2) . A chain of m causally ordered points requires proper time extent $\tau \geq m \cdot \rho^{-1/d} \sim \ell$ between q_{i_1} and q_{i_m} .

Since v is timelike, the projection $v \cdot \tilde{w}'_k$ is approximately the time coordinate of q_k relative to $\Phi(x)$. The endpoints of the chain satisfy $|v \cdot \tilde{w}'_{i_1} - v \cdot \tilde{w}'_{i_m}| \geq c \ell$ for a dimensional constant $c > 0$. Therefore at least one of $|v \cdot \tilde{w}'_{i_1}|, |v \cdot \tilde{w}'_{i_m}|$ is $\geq c \ell / 2$, contributing at least $c^2 \ell^2 / 4$ to the weighted variance (since the chain elements have weights $w_{i_j} = \Theta(1)$ from being within $\sim \ell$ of x). Hence $v^T \Sigma_2 v \geq c' \ell^2$ in the timelike direction.

Spacelike case. Now suppose v is spacelike. Suppose for contradiction that $v^T \Sigma_2 v = \varepsilon^2$ with $\varepsilon \ll \ell$. By Chebyshev's inequality applied to the weighted measure $d\mu = w_k d\delta_k / \sum w_k$:

$$\mu(\{k : |v \cdot \tilde{w}'_k| > \eta\}) \leq \frac{\varepsilon^2}{\eta^2} \quad \text{for any } \eta > 0.$$

Choose $\eta := \sqrt{\varepsilon \ell}$ (the geometric mean of ε and ℓ), so that $\varepsilon^2 / \eta^2 = \varepsilon / \ell \rightarrow 0$ as $\varepsilon \rightarrow 0$. Then a fraction $1 - \varepsilon / \ell$ of the total weight comes from points with $|v \cdot \tilde{w}'_k| \leq \sqrt{\varepsilon \ell}$, i.e. from points in a slab of width $\sqrt{\varepsilon \ell} \ll \ell$ perpendicular to v .

Apply (F2) to the causal diamond. Take source elements $c_a \prec c_b$ in C with both within distance $\sim \ell$ of x in M_1 and proper time $\tau_1(f_1(c_a), f_1(c_b)) \sim \ell$. By Proposition 3.6, the corresponding target proper time is also $\sim \ell$. The causal diamond $D_2 := J^+(f_2(c_a)) \cap J^-(f_2(c_b))$ in M_2 has h_2 -volume $\Theta(\ell^d)$, and by (F1) it contains the images of all $\Theta(\rho \ell^d)$ elements of the causal interval $I(c_a, c_b) \subset C$.

The intersection of D_2 with the slab of width $\sqrt{\varepsilon \ell}$ perpendicular to v has volume $O(\sqrt{\varepsilon \ell} \cdot \ell^{d-1})$ (slab cross-section times diamond extent in the other $d-1$ directions). By (F2) at scale ℓ :

$$|f_2(C) \cap D_2 \cap \text{slab}| \leq (1 + \delta_\ell) \rho \cdot O(\sqrt{\varepsilon \ell} \cdot \ell^{d-1}) = O(\rho \ell^{d-1/2} \sqrt{\varepsilon}).$$

But by Chebyshev, this region contains a fraction $(1 - \varepsilon/\ell)$ of the $\Theta(\rho\ell^d)$ points in D_2 , so:

$$(1 - \varepsilon/\ell)\rho\ell^d \leq O(\rho\ell^{d-1/2}\sqrt{\varepsilon}).$$

For $\varepsilon \ll \ell$, the left side is $\sim \rho\ell^d$, giving $\ell^{1/2} \lesssim \sqrt{\varepsilon}$, i.e. $\varepsilon \gtrsim \ell$ —a contradiction.

Both cases yield $v^T \Sigma_2 v \geq c_1 \ell^2$ for a dimensional constant $c_1 > 0$ depending on d and δ_ℓ . By the Rayleigh characterisation, $\sigma_{\min}(\Sigma_2) \geq c_1 \ell^2$. The uniform bound over all x follows by the same union-bound argument as in Lemma 3.15. $\square$

Proposition 3.17 (Φ is a diffeomorphism). *For ρ sufficiently large, Φ is a global diffeomorphism.*

Proof. Step 1: $D\Phi$ is invertible. The first-order condition for the Karcher mean (6) is

$$F(x, y) := \sum_k w_k(x) \exp_y^{-1}(q_k) = 0.$$

Implicit differentiation gives $D\Phi(x) = -[D_y F(\Phi(x))]^{-1} \circ D_x F(x)$.

The matrix $D_y F$ is exactly $\sum_k w_k(x) \partial_y [\exp_y^{-1}(q_k)]$, where ∂_y is the derivative with respect to the basepoint y at fixed q_k . Since the weights $w_k(x)$ depend only on x , not on y , taking the y -derivative hits only the inverse exponential map. In flat space, $\exp_y^{-1}(q_k) = q_k - y$ gives $\partial_y [\exp_y^{-1}(q_k)] = -I$ exactly. In curved space, the Jacobi-field expansion gives $\partial_y [\exp_y^{-1}(q_k)] = -I + O(|\tilde{w}'_k|^2/\lambda^2)$, where $\tilde{w}'_k = \exp_y^{-1}(q_k)$. Summing:

$$D_y F = -W[I + O(\ell^2/\lambda^2)],$$

where $W := \sum_k w_k$, since $|\tilde{w}'_k| = O(\ell)$ for points contributing significantly. Since $\ell \ll \lambda$, $D_y F$ is a small perturbation of $-WI$ and is strictly invertible: $[D_y F]^{-1} = -(1/W)[I + O(\ell^2/\lambda^2)]$.

The matrix $D_x F = \sum_k (Dw_k) \otimes \exp_y^{-1}(q_k)$ involves $Dw_k = +\frac{2}{\ell^2} w_k \tilde{w}_k$ (since $w_k = \exp(-|x - p_k|^2/\ell^2)$ and $\tilde{w}_k = p_k - x$). Therefore $D_x F = +\frac{2}{\ell^2} \sum_k w_k \tilde{w}'_k \otimes \tilde{w}_k$, which equals $(2W/\ell^2)\Sigma_{\text{cross}}$ where $\Sigma_{\text{cross}} = \frac{1}{W} \sum_k w_k \tilde{w}'_k \tilde{w}_k^T$ is the weighted cross-covariance. By Corollary 3.8, $\sigma_{\min}(\Sigma_{\text{cross}}) \geq c_2 \ell^2$, so $D_x F$ is invertible with $\sigma_{\min}(D_x F) \geq 2Wc_2$.

Therefore $D\Phi = -[D_y F]^{-1} D_x F$ is a product of two invertible operators, hence invertible, with $\sigma_{\min}(D\Phi) \geq c_0 > 0$.

Step 2: Constant sign of the Jacobian. Define the Jacobian $J_\Phi: M_1 \rightarrow \mathbb{R}$ coordinate-free by $\Phi^* \text{vol}_{h_2} = J_\Phi \cdot \text{vol}_{h_1}$, where vol_{h_i} are the Riemannian volume forms of the auxiliary metrics. By Step 1, $D\Phi$ is invertible everywhere, hence $J_\Phi \neq 0$ everywhere. Since J_Φ is continuous on the connected manifold M_1 , it has constant sign throughout. We do not at this stage determine which sign; this only matters for orientation, which we discuss in Remark 3.18 below.

Step 3: Global bijectivity.

Domain restriction (geometric). With cutoff weights, $\Phi(x)$ is defined only when at least one $w_k(x) > 0$, i.e., when the ball $B_{2\alpha\ell}^{h_1}(x)$ contains at least one embedded point p_k . To obtain a domain on which Φ is robustly defined and Hadamard–Caccioppoli applies, we define M_1° purely geometrically, independent of the random sprinkling:

$$M_1^\circ := \{x \in M_1 : d_{h_1}(x, \partial M_1) > 4\alpha\ell\}. \quad (28)$$

This is a deterministic open subset of M_1 with smooth (deterministic) boundary, contained strictly in the interior of M_1 at h_1 -distance $\geq 4\alpha\ell$ from ∂M_1 . The factor $4\alpha\ell$ is chosen so that the cutoff support $B_{2\alpha\ell}^{h_1}(x)$ of any $x \in M_1^\circ$ lies wholly inside M_1 at distance $\geq 2\alpha\ell$ from ∂M_1 .

Probabilistic density on M_1° . We must verify that for every $x \in M_1^\circ$, there exist embedded points p_k in the cutoff support. By Chernoff's inequality, for fixed $x \in M_1^\circ$, the count $|\{p_k \in B_\ell^{h_1}(x)\}|$ is Poisson with mean $\geq c_d \rho \ell^d$, so $\Pr(|\{p_k \in B_\ell^{h_1}(x)\}| < c_d \rho \ell^d / 2) \leq \exp(-c'_d \rho \ell^d)$. Take a ℓ -net $\mathcal{N} \subset M_1^\circ$ of size $|\mathcal{N}| \leq C \text{Vol}_{h_1}(M_1^\circ) / \ell^d$. Union bound over $\mathcal{N}$: with probability

$$\geq 1 - C(\text{Vol}(M_1) / \ell^d) \exp(-c'_d \rho \ell^d), \quad (29)$$

every net point has $\geq c_d \rho \ell^d / 2$ embedded points in its ℓ -ball. Between net points, the count is continuous (changes by at most the count in a symmetric difference of volume $O(\ell^d)$). Hence with probability (29), every $x \in M_1^\circ$ has at least $c_d \rho \ell^d / 4$ embedded points in $B_\ell^{h_1}(x) \subset \text{supp}(w(x, \cdot))$. In particular, $\Phi(x)$ is well-defined for every $x \in M_1^\circ$ on this high-probability event.

Target domain. Define $M_2^\circ \subset M_2$ analogously by $d_{h_2}(\cdot, \partial M_2) > 4\alpha\ell$. By Corollary 3.7, the image $\Phi(M_1^\circ)$ lies within h_2 -distance $O(\varepsilon_\tau \alpha\ell)$ of the corresponding region in M_2 , hence inside M_2 at distance $\geq 4\alpha\ell - O(\varepsilon_\tau \alpha\ell) \geq 3\alpha\ell$ from ∂M_2 for ρ large. Thus $\Phi(M_1^\circ) \subseteq M_2^\circ$.

Properness on M_1° . We show that $\Phi: M_1^\circ \rightarrow M_2^\circ$ is proper: the preimage of any compact set is compact. Let $K \subset M_2^\circ$ be compact. We show $\Phi^{-1}(K) \subset M_1^\circ$ is compact.

By the well-definedness argument (Proposition 3.5(i)), $\Phi(x)$ lies within the ball $B_{4\alpha\ell}^{h_2}(\bar{q}(x))$, where $\bar{q}(x) := \frac{1}{|S(x)|} \sum_{k \in S(x)} q_k$ is the unweighted average of active target points $S(x) := \{k : w_k(x) > 0\}$. Hence $\Phi(x) \in K$ implies $\bar{q}(x) \in K_{4\alpha\ell} := \{y \in M_2 : d_{h_2}(y, K) \leq 4\alpha\ell\}$ (still compact in M_2).

The active target points $\{q_k : k \in S(x)\}$ are within $2\alpha\ell$ of $\bar{q}(x)$ in h_2 -distance (by definition of unweighted average over a bounded set combined with Cor. 3.7), so they lie in $K' := K_{6\alpha\ell}$. Since $f_2: C \rightarrow M_2$ is injective, the corresponding source points $\{p_k : k \in S(x)\} = f_1(f_2^{-1}(\{q_k : k \in S(x)\}))$ are a finite subset of M_1 . By faithfulness (F2), the source points lie in a region of M_1 whose h_1 -volume is bounded by $|f_2^{-1}(K')| / \rho \leq (1 + \delta_{6\alpha\ell}) \text{Vol}_{h_2}(K') / \rho$, which is compact (it is contained in the closure of $f_1(f_2^{-1}(K'))$), a closed subset of the compact $K'' :=$ the closed $4\alpha\ell$ -neighbourhood of $f_1(f_2^{-1}(K'))$ in M_1 .

Finally, x is within $2\alpha\ell$ of some active source point $p_k \in K''$, so $x \in K''_{2\alpha\ell}$, a compact set. Therefore $\Phi^{-1}(K) \subset K''_{2\alpha\ell} \cap M_1^\circ$, a closed subset of a compact set, hence compact. This proves properness.

Hadamard–Caccioppoli. Φ is now a proper local diffeomorphism $M_1^\circ \rightarrow M_2^\circ$ between connected manifolds (with M_1°, M_2° open and connected for ρ sufficiently large). By the Hadamard–Caccioppoli global inversion theorem (Krantz–Parks [11], Thm. 6.2.8), Φ is a covering map onto its image. Since M_2° is connected, the image is a clopen subset of M_2° , hence either empty or all of M_2° ; since Φ maps embedded p_k to nearby $q_k \in M_2^\circ$, the image is nonempty, hence onto. Thus $\Phi: M_1^\circ \rightarrow M_2^\circ$ is a surjective covering map of constant degree $k \in \mathbb{Z}_{>0}$.

Degree-one argument. We compute k by computing the cardinality of one fiber. Fix any $x_0 \in M_1^\circ$ and let $y_0 := \Phi(x_0) \in M_2^\circ$. By Corollary 3.8 (the triliteration result), on the inner-core scale around x_0 the map Φ satisfies $\Phi(x_0 + v) = y_0 + \hat{\Lambda}v + R(v)$ for any v with $|v|_{h_1} \leq 2\ell$, with $\hat{\Lambda} \in O(1, d-1)$ and $\|R(v)\| \leq C_d \varepsilon_\tau \alpha\ell$. By shrinking the perturbation scale:

Φ restricted to $B_{2\ell}^{h_1}(x_0)$ is a small C^1 -perturbation of the linear map $v \mapsto \hat{\Lambda}v$, which has degree $\det(\hat{\Lambda}) = \pm 1$ as an element of $O(d)$ (here we consider $\hat{\Lambda}$ as a $d \times d$ matrix, irrespective of its Lorentz structure).

Hence the local degree of Φ at x_0 (i.e., the contribution of x_0 to the algebraic count of preimages of y_0) is ± 1 . By Step 2 (constant sign of J_Φ), all local degrees have the same sign. The fiber $\Phi^{-1}(y_0)$ has cardinality equal to the algebraic degree $|k|$, but each preimage contributes ± 1 with the same sign (constant Jacobian sign), so $k \geq 1$.

To get $k \leq 1$: suppose for contradiction that there are two distinct preimages $x_0, x'_0 \in \Phi^{-1}(y_0)$. Consider the two local trilateration linear models around x_0 and x'_0 , with Lorentz transformations $\hat{\Lambda}$ and $\hat{\Lambda}'$. Both map a neighbourhood of x_0 (resp. x'_0) close to a neighbourhood of y_0 . Now, in the inner-core trilateration, the anchor points and the resulting $\hat{\Lambda}$ are determined (up to errors $O(\varepsilon_\tau \alpha \ell)$) by the embedded points near x_0 . Specifically, $\hat{\Lambda}$ maps source displacement vectors $p_k - x_0$ (for embedded p_k near x_0) to target displacement vectors $q_k - y_0$ (for the images, which are embedded q_k near y_0).

If $x'_0 \neq x_0$ but $\Phi(x'_0) = \Phi(x_0) = y_0$, the same embedded q_k near y_0 would be reached from two distinct centres in M_1° . But each q_k corresponds under $f_2 \circ f_1^{-1}$ to a unique source point $p_k \in f_1(C)$, so the inner-core anchors around x_0 and around x'_0 would share images, forcing $\|x_0 - x'_0\| \leq O(\varepsilon_\tau \alpha \ell) + O(\ell)$ (a constant fraction of the inner-core radius). This contradicts $x_0 \neq x'_0$ at separation $\gg \ell$.

For separations $\|x_0 - x'_0\| = O(\ell)$, the same argument with the Karcher mean's local uniqueness (Proposition 3.5(i)) gives a contradiction: $\Phi(x_0) = \Phi(x'_0)$ would require two distinct minima of the Karcher objective, but the objective is strictly convex in the relevant target ball.

Hence $k = 1$, and $\Phi : M_1^\circ \rightarrow M_2^\circ$ is a single-sheeted covering, i.e., a diffeomorphism.

Extension to M_1 . For compact M_1 , the complement $M_1 \setminus M_1^\circ$ is a thin ℓ -neighbourhood of ∂M_1 which shrinks to a measure-zero boundary as $\rho \rightarrow \infty$ (equivalently $\ell \rightarrow 0$). The diffeomorphism extends to this boundary region by continuity. For noncompact M_1 , the construction applies to any precompact subdomain and the conclusion is that Φ is a diffeomorphism in the interior, which is the physically meaningful statement. $\square$

Remark 3.18 (Orientation). The conclusion “ Φ is a diffeomorphism” allows Φ to be either orientation-preserving or orientation-reversing relative to fixed orientations on M_1 and M_2 . Both possibilities are consistent with the faithful embedding hypotheses: a global spatial reflection in $d \geq 4$ preserves Lorentzian distances, causal order (F1), and volumes (F2), so two faithful embeddings related by such a reflection are equally valid. The Hauptvermutung therefore concludes that M_1 and M_2 are approximately isometric *as unoriented Lorentzian manifolds*. If one assumes in addition that the two faithful embeddings are orientation-compatible (a natural condition when both spacetimes carry fixed physical orientations), the diffeomorphism Φ is automatically orientation-preserving: the local correspondence $\tilde{w}_k \mapsto \tilde{w}'_k$ inherits orientation compatibility, hence $\det(\Sigma_{\text{cross}}) > 0$ at each embedded point, hence $J_\Phi > 0$ globally by Step 2.

Remark 3.19 (No exotic $\mathbb{R}^4$ obstruction). The construction is entirely local: at each x , the Karcher mean uses only nearby point correspondences. No topological classification of manifolds is invoked, so the proof works uniformly in all dimensions including $d = 4$ [7, 8].

Remark 3.20 (What this argument avoids). The invertibility proof uses only:

- Poisson isotropy $\Rightarrow \Sigma_1$ non-degenerate (probability),
- causal non-collapse $\Rightarrow \Sigma_2$ non-degenerate (combinatorics of partial orders),
- implicit function theorem $\Rightarrow D\Phi$ invertible (calculus).

No “finite-point Malament” theorem is used. We never identify a Lorentz transformation, best-fit linear map, or any geometric relationship between the two manifolds. The non-degeneracy of Σ_2 is a purely *combinatorial* consequence of the richness of the causal order, not a metric statement.

4 Conformal rigidity

We now show that Φ is an approximate conformal isometry, using the distance preservation result established in Section 3.4.

4.1 From distance preservation to conformal isometry

Lemma 4.1 (Approximate Alexandrov–Zeeman). *Let $A: \mathbb{R}^{1,d-1} \rightarrow \mathbb{R}^{1,d-1}$ be an invertible linear map. If A maps the null cone $\mathcal{N} = \{v : \eta(v, v) = 0\}$ to within angular error $\theta \ll 1$ (i.e., for every null v , $|\eta(Av, Av)|/|Av|^2 \leq \theta$), then there exist $\Omega > 0$ and $\Lambda \in O(1, d-1)$ with $\|A - \Omega\Lambda\| \leq C_d \|A\| \theta$, where C_d depends only on the dimension d .*

Intuition. The condition that A approximately preserves the Minkowski null cone forces $A^T \eta A$ to be approximately proportional to η (an exact equality $A^T \eta A = \Omega^2 \eta$ characterises scaled Lorentz transformations). Approximate equality then forces A to be close to such a transformation by stability of the orthogonal group within $GL(d)$.

(A naive “singular value” argument fails here: a pure Lorentz boost preserves the null cone exactly but has singular values $e^{\pm \xi}$ with arbitrary disparity, because the SVD diagonalises with respect to the Euclidean metric, not the Minkowski metric. The argument below works intrinsically with the quadratic form $A^T \eta A$ and avoids the Euclidean trap.)

Proof. Step 1: $A^T \eta A$ is approximately proportional to η . Define the symmetric quadratic form $Q := A^T \eta A$. The hypothesis on A gives, for every null $v \in \mathcal{N}$,

$$|v^T Q v| = |\eta(Av, Av)| \leq \theta |Av|^2 \leq \theta \|A\|^2 |v|^2.$$

We claim that any symmetric form Q on $\mathbb{R}^d$ satisfying $|v^T Q v| \leq \beta |v|^2$ for all null v is close to a multiple of η :

$$Q = \alpha \eta + R, \quad \|R\| \leq C_d \beta, \quad (30)$$

for some scalar $\alpha \in \mathbb{R}$ and a dimensional constant C_d .

To see this, parametrise null vectors as $v = (1, \hat{n})$ with $\hat{n} \in S^{d-2}$. Then $v^T Q v = Q_{00} + 2Q_{0i} \hat{n}^i + Q_{ij} \hat{n}^i \hat{n}^j$. Replacing $\hat{n} \rightarrow -\hat{n}$ (also null) gives $Q_{00} - 2Q_{0i} \hat{n}^i + Q_{ij} \hat{n}^i \hat{n}^j$. Subtracting: $|4Q_{0i} \hat{n}^i| \leq 2\beta \cdot 2$ for all $\hat{n} \in S^{d-2}$, hence $\|Q_{0i}\| \leq \beta$ for each spatial i . Adding the two: $|2Q_{00} + 2Q_{ij} \hat{n}^i \hat{n}^j| \leq 4\beta$, so $|Q_{ij} \hat{n}^i \hat{n}^j + Q_{00}| \leq 2\beta$ for every unit $\hat{n}$. The trace-free part of Q_{ij} thus has operator norm $\leq C\beta$ on the unit sphere, and the trace satisfies $|\text{tr}(Q_{ij})/(d-1) + Q_{00}| \leq 2\beta$.

Set $\alpha := -Q_{00}$ (so that $\alpha \eta_{00} = Q_{00}$). Then the residual $R := Q - \alpha \eta$ has $R_{00} = 0$ exactly, $|R_{0i}| \leq \beta$, and $R_{ij} = Q_{ij} - \alpha \delta_{ij} = Q_{ij} + Q_{00} \delta_{ij}$ satisfies $|R_{ij} \hat{n}^i \hat{n}^j| \leq 2\beta$ for all unit $\hat{n}$. Since

the diagonal entries of R_{ij} are bounded (by setting $\hat{n} = e_i$) and the off-diagonal entries are bounded by polarisation, $\|R_{ij}\| \leq C\beta$. Hence $\|R\| \leq C_d\beta$ as claimed.

Applying this to our $Q = A^T\eta A$ with $\beta = \theta\|A\|^2$:

$$A^T\eta A = \Omega^2\eta + R, \quad \|R\| \leq C_d\theta\|A\|^2, \quad (31)$$

where we have written $\alpha = \Omega^2$ for some real Ω (the sign of α will be confirmed positive below, since A is invertible and the cone condition forces $\alpha \neq 0$).

Step 2: $\Omega > 0$ and A is close to Ω times a Lorentz transformation. Take any timelike vector v_0 (so $\eta(v_0, v_0) < 0$). By (31), $\eta(Av_0, Av_0) = \Omega^2\eta(v_0, v_0) + v_0^T R v_0$. The left side has the sign of $\eta(v_0, v_0)$ (since A is near $\Omega\Lambda$, an approximate Lorentz transformation, which preserves causality up to error θ). Comparing signs forces $\Omega^2 > 0$, hence $\Omega \in \mathbb{R}$.

Define $B := \Omega^{-1}A$. Then $B^T\eta B = \Omega^{-2}A^T\eta A = \eta + \Omega^{-2}R$, so

$$\|B^T\eta B - \eta\| \leq C_d\theta\|A\|^2/\Omega^2 = O(\theta).$$

(The ratio $\|A\|/\Omega$ is bounded by a dimensional constant: for a vector v with $\eta(v, v) \neq 0$, the estimate (31) gives $|\eta(Av, Av) - \Omega^2\eta(v, v)| \leq C\theta\|A\|^2|v|^2$, so $\Omega^2 \geq c\|A\|^2$ for any unit v with $|\eta(v, v)|$ bounded below.)

Step 3: Stability of the Lorentz group. The map $\Psi: GL(d, \mathbb{R}) \rightarrow \text{Sym}(d)$ defined by $\Psi(B) := B^T\eta B$ is smooth, and $\Psi^{-1}(\eta) = O(1, d-1)$. The differential of Ψ at $\Lambda \in O(1, d-1)$ is $d\Psi_\Lambda(E) = \Lambda^T\eta E + E^T\eta\Lambda$, which has full rank onto its image (the tangent space to the orbit of η under congruence). By the implicit function theorem applied near $\Lambda = I$, for B satisfying $\|B^T\eta B - \eta\| \leq \delta$ with δ sufficiently small, there exists $\Lambda \in O(1, d-1)$ with $\|B - \Lambda\| \leq C_d\delta$. Applying this with $\delta = O(\theta)$ from Step 2:

$$\|\Omega^{-1}A - \Lambda\| = \|B - \Lambda\| = O(\theta),$$

hence $\|A - \Omega\Lambda\| = \Omega \cdot O(\theta) = O(\|A\|\theta)$. $\square$

Proposition 4.2 (Approximate conformal isometry). *The diffeomorphism Φ satisfies $\Phi^*g_2 = \Omega^2g_1 + E$ where $\Omega > 0$ is smooth and $\|E\|_{g_1} = O(\varepsilon_\tau + \ell^2/\lambda^2 + (\rho\ell^d)^{-1/2})$.*

Proof. The argument uses the trilateration result (Corollary 3.8) to identify $D\Phi$ with an approximate Lorentz transformation, then applies Lemma 4.1 to extract the conformal isometry.

Step 1: $D\Phi$ via the cross-covariance. By Proposition 3.5(iv), the empirical derivative satisfies $D\Phi = -[D_y F]^{-1}D_x F = (2/\ell^2)\Sigma_{\text{cross}} + O(\ell^2/\lambda^2)$, where the $O(\ell^2/\lambda^2)$ correction is from the curvature of the exponential map in $[D_y F]^{-1}$. The cross-covariance Σ_{cross} has already been analysed in Corollary 3.8: on the high-probability event of the trilateration argument,

$$\Sigma_{\text{cross}}^{\text{core}} = \hat{\Lambda}\Sigma_1^{\text{core}} + E^{\text{core}}, \quad \|E^{\text{core}}\| = O(\varepsilon_\tau\alpha\ell^2), \quad (32)$$

where $\hat{\Lambda} \in O(1, d-1)$ is the Lorentz transformation arising from the anchor scaffold (Lemma 3.12) and $\Sigma_1^{\text{core}} = \frac{1}{W^{\text{core}}} \sum_{|p_k| \leq 2\ell} w_k p_k p_k^T$ is the inner-core source covariance.

Source covariance is isotropic. By Lemma 3.15 applied to the inner core, $\Sigma_1^{\text{core}} = (\ell^2/2)I + E_1$ with $\|E_1\| = O(\ell^2/\sqrt{N_{\text{eff}}})$ where $N_{\text{eff}} := \rho\ell^d$ is the effective Poisson sample size on the inner core (matrix Bernstein). Substituting:

$$\Sigma_{\text{cross}}^{\text{core}} = \frac{\ell^2}{2}\hat{\Lambda} + \hat{\Lambda}E_1 + E^{\text{core}} = \frac{\ell^2}{2}\hat{\Lambda} + O(\ell^2/\sqrt{N_{\text{eff}}} + \varepsilon_\tau\alpha\ell^2). \quad (33)$$

Extension to full Σ_{cross} . The full cross-covariance differs from the inner-core $\Sigma_{\text{cross}}^{\text{core}}$ by the boundary-layer contribution analysed in Step G of the proof of Corollary 3.8: with weight fraction $\beta \geq 0.9$ for the inner core (Gaussian tail estimate), and boundary contribution $\|\Sigma_{\text{cross}}^{\text{bdry}}\| \leq C'\ell^2$ (controlled by the exponentially small weights at $|p| \geq 2\ell$),

$$\Sigma_{\text{cross}} = \beta \Sigma_{\text{cross}}^{\text{core}} + (1 - \beta) \Sigma_{\text{cross}}^{\text{bdry}} = \frac{\ell^2}{2} \hat{\Lambda} + O(\ell^2 / \sqrt{N_{\text{eff}}} + \varepsilon_\tau \alpha \ell^2).$$

Combining with $D\Phi = (2/\ell^2) \Sigma_{\text{cross}} + O(\ell^2/\lambda^2)$:

$$D\Phi = \hat{\Lambda} + O(1/\sqrt{N_{\text{eff}}} + \varepsilon_\tau \alpha + \ell^2/\lambda^2). \quad (34)$$

Null cone preservation. Since $\hat{\Lambda} \in O(1, d-1)$, it preserves the null cone exactly: $\eta(\hat{\Lambda}v, \hat{\Lambda}v) = 0$ for null v . For $D\Phi = \hat{\Lambda} + E$ with $\|E\| = O(\varepsilon_\tau \alpha + \ell^2/\lambda^2 + (\rho\ell^d)^{-1/2})$:

$$|\eta(D\Phi \cdot v, D\Phi \cdot v)| = |\eta(\hat{\Lambda}v + Ev, \hat{\Lambda}v + Ev)| \leq 2\|\eta\| \cdot \|\hat{\Lambda}v\| \cdot \|Ev\| + \|\eta\| \cdot \|Ev\|^2 = O(\|E\| \|v\|^2).$$

Hence $D\Phi$ maps the null cone to within angular error $\theta = O(\|E\|) = O(\varepsilon_\tau \alpha + \ell^2/\lambda^2 + (\rho\ell^d)^{-1/2})$.

Step 2: Approximate Alexandrov–Zeeman. By Step 1, $D\Phi(x)$ maps the null cone at x to within angular error $\theta = O(\varepsilon_\tau \alpha + \ell^2/\lambda^2 + (\rho\ell^d)^{-1/2})$ of the null cone at $\Phi(x)$. By Lemma 4.1, $D\Phi(x) = \Omega(x)\Lambda(x) + O(\theta)$ for $\Omega(x) > 0$ and $\Lambda(x) \in O(1, d-1)$. Hence $\Phi^*g_2 = \Omega^2g_1 + E$ with $\|E\| = O(\theta)$. $\square$

5 Volume rigidity

Proposition 5.1 (Volume rigidity). *Under $\Phi^*g_2 = \Omega^2g_1 + E$ with $\|E\|$ small, the conformal factor satisfies*

$$|\Omega(p) - 1| = O(\rho^{-1/(d+2)} \lambda^{-d/(d+2)} \sqrt{\log \rho V_M}).$$

Remark 5.2 (Interpretation of the rate). The bound reflects two ingredients: (a) the curvature error $O(\ell_D/\lambda)$ from the local-flatness approximation of g within the diamond, and (b) the F2 tolerance $\delta_{\ell_D} = K_d \sqrt{\log \rho V_M / (\rho \ell_D^d)}$ at the optimised scale $\ell_D = \rho^{-1/(d+2)} \lambda^{2/(d+2)}$. The optimisation balances these exactly. The $\sqrt{\log \rho V_M}$ factor is the union-bound cost of requiring F2 to hold simultaneously over all admissible diamonds and is irreducible.

Proof of Proposition 5.1. Under $\Phi^*g_2 \approx \Omega^2g_1$, volumes transform as $\text{Vol}_{g_2}(\Phi(D)) \approx \int_D \Omega^d dV_{g_1}$. Choose a diamond D of diameter $\ell_D \ll \lambda$ centred at p . With the scale-dependent tolerance from Definition 2.3 (F2), each embedding gives:

$$\frac{|f_i(C) \cap D_i|}{\rho} = \text{Vol}_{g_i}(D_i)(1 + O(\delta_D)), \quad \delta_D = K_d \sqrt{\frac{\log \rho V_M}{\rho \text{Vol}_g(D)}}.$$

Both counts equal $|C \cap D|$ since Φ matches embedded points (Proposition 3.5(iii)), so $\text{Vol}_{g_1}(D) \approx \text{Vol}_{g_2}(\Phi(D))$ up to relative error $O(\delta_D)$. Hence

$$\frac{1}{V} \int_D \Omega^d dV_{g_1} = 1 + O(\delta_D).$$

To bound Ω^d pointwise, observe that inside D the smooth conformal factor varies by at most $O(\ell_D/\lambda)$: $|\Omega^d(p) - \langle \Omega^d \rangle_D| \leq C\ell_D/\lambda$. The Poisson noise in the count contributes the δ_D term, which already encodes the $\sqrt{1/(\rho\ell_D^d)}$ scaling. Combining:

$$|\Omega^d(p) - 1| \leq C_1 \frac{\ell_D}{\lambda} + C_2 \delta_D = C_1 \frac{\ell_D}{\lambda} + C_2 K_d \sqrt{\frac{\log \rho V_M}{\rho \ell_D^d}}. \quad (35)$$

Optimisation of ℓ_D . Balancing the two terms: $\ell_D/\lambda = \sqrt{\log \rho V_M / (\rho \ell_D^d)}$, i.e. $\ell_D^{d+2} = \lambda^2 \rho^{-1} \log \rho V_M$, giving

$$\ell_D = \lambda^{2/(d+2)} \rho^{-1/(d+2)} (\log \rho V_M)^{1/(d+2)}.$$

At this optimum,

$$|\Omega^d - 1| \leq O(\rho^{-1/(d+2)} \lambda^{-d/(d+2)} (\log \rho V_M)^{1/(d+2)}).$$

For typographic simplicity, we absorb the $(\log \rho V_M)^{1/(d+2)}$ into a $\sqrt{\log \rho V_M}$ factor (which it is dominated by), yielding the stated bound

$$|\Omega - 1| = O(\rho^{-1/(d+2)} \lambda^{-d/(d+2)} \sqrt{\log \rho V_M}).$$

Since $\Omega > 0$ and $\Omega^d - 1 = d(\Omega - 1) + O((\Omega - 1)^2)$ for Ω near 1, the bound on $|\Omega^d - 1|$ transfers to $|\Omega - 1|$.

Verifying F2 applicability at ℓ_D . The optimal ℓ_D satisfies $\ell_D \ll \lambda$ iff $\rho \lambda^d \gg \log \rho V_M$ (which is implied by the scale separation hypothesis). Similarly $\ell_D \gg \tau_{\min}$ iff $\rho^{2/(d(d+2))} \lambda^{2/(d+2)} \gg (\log \rho V_M)^{2/d-1/(d+2)}$, which holds for sufficiently large ρ (power of ρ beats logs). Hence $\ell_D \in [\tau_{\min}, c_* \lambda]$ for sufficiently large ρ , and F2 applies directly at scale ℓ_D with the scale-dependent tolerance δ_{ℓ_D} . $\square$

6 Main theorem

Theorem 6.1 (The Hauptvermutung, deterministic form). *Let C be a finite causal set with two $(\rho; c_*, K_d)$ -faithful embeddings $f_1: C \hookrightarrow (M_1, g_1)$ and $f_2: C \hookrightarrow (M_2, g_2)$ (Definition 2.3) into globally hyperbolic d -dimensional Lorentzian manifolds with curvature scale λ , satisfying $\rho \lambda^d \geq c_*^{-2d} (\log \rho V_{\max})^2$. Assume additionally that the matrix-concentration event $\mathcal{E}_{\text{mat}}$ holds (Lemma 3.15, Corollary 3.8), i.e., the Poisson-derived source covariance and cross-covariance bounds hold uniformly on the smoothing-scale net.*

Then on the open subdomains M_1°, M_2° (Equation (28)), there exists a diffeomorphism $\Phi: M_1^\circ \rightarrow M_2^\circ$ that is an ε -approximate isometry with

$$\varepsilon = \underbrace{O(\rho^{-1/(d+2)} \lambda^{-d/(d+2)} \sqrt{\log \rho V_{\max}})}_{\text{volume rigidity (incl. F2 at } \ell_D)} + \underbrace{O((\rho \ell_D^d)^{-\alpha_d} + (\rho \ell_D^d)^{-1/2} + \ell_D^2/\lambda^2)}_{\text{conformal rigidity}}. \quad (36)$$

Here the F2 tolerance at scale ℓ_D is $\delta_{\ell_D} = K_d \sqrt{\log \rho V_{\max} / (\rho \ell_D^d)} = O(\rho^{-1/(d+2)} \lambda^{-d/(2(d+2))} \sqrt{\log \rho V_{\max}})$, already absorbed into the volume-rigidity term.

Optimising the smoothing scale ℓ over the conformal contributions, using the rigorous BB convergence rate $\alpha_d = 1/(2d)$ from [4] (Theorems 3 and 9), gives the optimum at $\ell_* = \rho^{-1/(5d)} \lambda^{4/5}$, with

$$\varepsilon = O\left(\rho^{-1/(d+2)} \lambda^{-d/(d+2)} \sqrt{\log \rho V_{\max}} + \rho^{-2/(5d)} \lambda^{-2/5} \log^{3/2}(\rho V_{\max})\right), \quad (37)$$

where $V_{\max} := \max(\text{Vol}_{g_1}(M_1), \text{Vol}_{g_2}(M_2))$. Since $2/(5d) < 1/(d+2)$ for all $d \geq 2$, the conformal contribution dominates the volume rigidity, giving

$$\varepsilon = O\left(\rho^{-2/(5d)} \lambda^{-2/5} \log^{3/2}(\rho V_{\max})\right). \quad (38)$$

Corollary 6.2 (The Hauptvermutung, probabilistic form). *Let (M_1, g_1) and (M_2, g_2) be globally hyperbolic d -dimensional Lorentzian manifolds with curvature scale λ and finite g -volumes $V_{M_1}, V_{M_2} < \infty$. Fix $\rho > 0$ with $\rho \lambda^d \geq c_*^{-2d} (\log \rho V_{\max})^2$. Let $\Pi_i \subset M_i$ ($i = 1, 2$) be independent Poisson processes of intensity $\rho \cdot dV_{g_i}$, and suppose there exists an order-isomorphism $\sigma: (\Pi_1, \preceq_1) \rightarrow (\Pi_2, \preceq_2)$ (i.e., a bijection of the sprinkled sets that respects causal order). Set $C := \Pi_1$ with $f_1 = \text{id}$ and $f_2 := \sigma$.*

Then with probability $\geq 1 - (\rho V_{\max})^{-K'_d}$ over the joint Poisson sprinkling, the inclusions f_1, f_2 are $(\rho; c_, K_d)$ -faithful embeddings (with K_d chosen sufficiently large; Lemma 2.6) and the matrix-concentration event $\mathcal{E}_{\text{mat}}$ holds, so the conclusion of Theorem 6.1 applies. In particular, with this probability there exists a diffeomorphism $\Phi: M_1^\circ \rightarrow M_2^\circ$ with ε -approximate isometry bound (37).*

In the limit $\rho \rightarrow \infty$ with λ fixed, $\varepsilon \rightarrow 0$ and the failure probability $\rightarrow 0$.

Remark 6.3 (Logical structure: deterministic vs probabilistic). Theorem 6.1 is purely deterministic: given two faithful embeddings of the same causal set, the conclusion follows without further randomness. The matrix-concentration event $\mathcal{E}_{\text{mat}}$ is part of the hypothesis, treated deterministically.

Corollary 6.2 provides the probabilistic realisation. The randomness is over the Poisson sprinkling that generates C and the induced embeddings, not over a “fixed” C (which would be incoherent: once C is fixed, no randomness remains). Lemma 2.6 guarantees the F2 hypothesis with high probability; Lemma 3.15 and Corollary 3.8 give the matrix-concentration event $\mathcal{E}_{\text{mat}}$.

This split addresses the common pitfall of mixing deterministic and probabilistic quantifiers in a single “theorem”. Both forms are useful: the deterministic form makes clear the logical structure (faithful embedding $\implies$ approximate isometry); the probabilistic form makes clear the physical relevance (sprinkling $\implies$ faithful embedding $\implies$ approximate isometry, with quantitative probability).

Remark 6.4 (On the optimal smoothing scale). Balancing ℓ^2/λ^2 against $(\rho \ell^d)^{-\alpha_d}$ gives $\ell_* = \rho^{-\alpha_d/(2+d\alpha_d)} \lambda^{2/(2+d\alpha_d)}$, with corresponding error $\rho^{-2\alpha_d/(2+d\alpha_d)} \lambda^{-2d\alpha_d/(2+d\alpha_d)}$. With the rigorously established $\alpha_d = 1/(2d)$ from Bollobás–Brightwell [4], $2+d\alpha_d = 5/2$, $2\alpha_d/(2+d\alpha_d) = 2/(5d)$, and $2d\alpha_d/(2+d\alpha_d) = 2/5$. The conformal error therefore scales as $\rho^{-2/(5d)} \lambda^{-2/5} \log^{3/2}(\rho V)$. Comparing exponents: $2/(5d) < 1/(d+2)$ for all $d \geq 2$, so the conformal contribution dominates the volume rigidity contribution in the asymptotic limit $\rho \rightarrow \infty$. For $d = 4$ this gives $\rho^{-1/10} \approx \rho^{-0.1}$ for the conformal term versus $\rho^{-1/6} \approx \rho^{-0.167}$ for volume rigidity: the conformal term is the slower-converging dominant error. A faster rate ($\alpha_d > 1/d$) would reverse this, and is the subject of conjecture (in the spirit of Conjecture 10 of [4]) but currently unproven in $d \geq 3$.

Proof of Theorem 6.1. The moving Karcher mean (Construction 3.2) gives a smooth map Φ on M_1° , with $D\Phi$ everywhere invertible by the cross-covariance non-degeneracy (Corollary 3.8, which uses the matrix-concentration event $\mathcal{E}_{\text{mat}}$). Proposition 3.17 establishes that Φ is a proper local diffeomorphism, hence a covering map by Hadamard–Caccioppoli; the degree argument (Remark 3.18, refined below) gives $|\deg(\Phi)| = 1$, making Φ a global diffeomorphism.

Proposition 4.2 gives $\Phi^*g_2 = \Omega^2g_1 + E$ with $\|E\| = O(\varepsilon_\tau + \ell^2/\lambda^2 + (\rho\ell^d)^{-1/2})$. Proposition 5.1 (with hypothesis F2 holding on the optimised scale ℓ_D , which lies in the admissible range $[c_*^{-1}\rho^{-1/d}(\log \rho V_M)^{2/d}, c_*\lambda]$ for sufficiently large ρ) gives $|\Omega - 1| = O(\rho^{-1/(d+2)}\lambda^{-d/(d+2)} + \delta)$. Combining and using $|\Omega^2 - 1| = 2|\Omega - 1| + O(|\Omega - 1|^2)$ yields (36).

The optimisation in Remark 6.4 yields (37) and (38). $\square$

Proof of Corollary 6.2. By Lemma 2.6 applied to each sprinkling, the inclusions f_1, f_2 are $(\rho, \delta; c_*)$ -faithful with probability $\geq 1 - 2(\rho V_{\max})^{-K'_d}$. Independently, Lemma 3.15 and Corollary 3.8 give the matrix-concentration event $\mathcal{E}_{\text{mat}}$ with probability $\geq 1 - (\rho V_{\max})^{-K''_d}$. Union bound, then Theorem 6.1. $\square$

7 Discussion

7.1 Proof architecture: order + number = geometry

The proof directly implements the CST doctrine:

- **Order** $\rightarrow$ causal non-collapse of the target (Lemma 3.16), longest-chain distance (Proposition 3.6), cone preservation (Proposition 4.2).
- **Number** $\rightarrow$ source covariance isotropy (Lemma 3.15), volume rigidity (Proposition 5.1).

7.2 Status of the arguments

The three stages of the proof rely on the following ingredients, which we classify by their current rigour.

Rigorous (provable from known results)

- (a) **Karcher mean smoothness and derivative bounds** (Proposition 3.5). Follows from Karcher [9] and the implicit function theorem.
- (b) **Source covariance non-degeneracy** (Lemma 3.15). Standard concentration inequality for Poisson processes.
- (c) **Target covariance non-degeneracy** (Lemma 3.16). Follows from the uniform density condition (F2) via a volume contradiction argument: $\rho\ell^d$ points cannot fit into a region of volume $\varepsilon\ell^{d-1}$ without violating (F2).
- (d) **Mesoscopic longest chain** (Proposition 3.6). The geometric sandwiching argument $D_{\eta^-} \subseteq D_g \subseteq D_{\eta^+}$ gives a deterministic squeeze $\ell_{C_{\eta^-}} \leq \ell_{C_g} \leq \ell_{C_{\eta^+}}$ on chain lengths (the same Poisson points are used; only the partial order changes, and chains in the more restrictive order are automatically chains in the less restrictive order). The bounding

diamonds $D_{\eta\pm}$ are rescaled standard Minkowski diamonds, so the Bollobás–Brightwell [4] estimates apply directly with effective volumes $V_{\pm} = V_g(1 \pm O(\epsilon))$.

- (e) **Cross-covariance non-degeneracy via trilateration** (Corollary 3.8). The discrete Procrustes argument combines four sub-lemmas: the trilateration identity (algebraic), anchor existence (Poisson concentration), finite Lorentzian Procrustes (reduces to Lemma 4.1), and trilateration extension (linear algebra). This bypasses the absence of spacelike distance estimators by triangulating all inner-core points via timelike distances to a fixed anchor scaffold.
- (f) **Approximate Alexandrov–Zeeman** (Lemma 4.1). Quadratic-form stability argument: a symmetric form vanishing on the null cone is a multiple of η , so approximate vanishing forces $A^T \eta A \approx \Omega^2 \eta$, then implicit function theorem applied to the smooth map $B \mapsto B^T \eta B$ near $O(1, d-1)$.
- (g) **Volume rigidity** (Proposition 5.1). Counting argument plus Jensen’s inequality, with explicit optimisation $\ell_D = \lambda^{2/(d+2)} \rho^{-1/(d+2)}$.
- (h) **No exotic $\mathbb{R}^4$ obstruction** (Remark 3.19). The construction never uses the topological classification of manifolds.
- (i) **Constant sign of the Jacobian** (Proposition 3.17, Step 2). Continuity of $J_{\Phi} \neq 0$ on the connected manifold M_1 . Whether the sign is positive or negative depends on whether the two embeddings are orientation-compatible (Remark 3.18); the sign is fixed only under such an additional hypothesis.

Remaining open

- (a) **Sharper convergence rate for longest chains in higher dimensions.** Our error bound depends on how fast the longest chain length h in a Poisson-sprinkled Minkowski causal diamond of volume V at density ρ converges to its asymptotic value $m_d(\rho V)^{1/d}$. Three facts about this convergence are relevant:

(i) *Leading-order limit*: Bollobás and Winkler [5] prove that the constant m_d exists, satisfying $1.77 \leq m_d \leq 2.62$ for $d \geq 3$ and $m_2 = 2$ exactly.

(ii) *Concentration around the mean*: Bollobás and Brightwell [4] (Theorem 3) prove that the longest chain length in the Poisson model ($H_{n,d}$ in their notation, with $n = \rho V$) deviates from its mean by at most $\lambda K_d(\rho V)^{1/(2d)} \log(\rho V) / \log \log(\rho V)$ with probability $\geq 1 - 4\lambda^2 e^{-\lambda^2}$ for appropriate λ .

(iii) *Rate of mean convergence*: They also prove (Theorem 9) that $|\langle H_{n,d} \rangle - m_d(\rho V)^{1/d}| \leq C(\rho V)^{1/(2d)} \log^{3/2}(\rho V) / \log \log(\rho V)$.

Combining (ii) and (iii) gives the relative error bound $|h/(m_d(\rho V)^{1/d}) - 1| = O((\rho V)^{-1/(2d)} \log^{3/2}(\rho V))$, i.e. a convergence rate of $1/(2d)$ up to logarithmic factors. This is the rate that enters our error bound $\rho^{-2/(5d)} \lambda^{-2/5}$ in (38).

The open problem is whether the exponent $1/(2d)$ is sharp. In $d = 2$, the analogous longest increasing subsequence problem is known to have fluctuations of order $n^{1/3}$ (Tracy–Widom [1]), giving a relative error $n^{-1/3+1/2} = n^{-1/6}$ rather than $n^{-1/4}$. So the BB

bound $1/(2d) = 1/4$ for $d = 2$ is not sharp; the true rate is $1/3$. Whether analogous improvements hold for $d \geq 3$ is conjectured but unproved (cf. Conjecture 10 of [4]). A proof of $\alpha_d > 1/d$ in $d \geq 3$ would directly sharpen our bound from (38) to one of the form $O(\rho^{-1/(d+2)} \lambda^{-d/(d+2)})$ (volume-rigidity rate). This is a question in geometric probability, independent of our work.

On the BB notation. The BB paper distinguishes the Poisson model (random number of points, $H_{n,d}$) from the binomial model (exactly n points iid uniform, $L_{n,d}$). The two have the same asymptotic constant m_d and the same rate exponent $1/(2d)$, with constants that differ by a factor of order one (their Theorem 8 vs Theorem 3). For sprinklings in CST, the Poisson version $H_{n,d}$ is the natural object, and that is what we use.

- (b) **Trilateration boundary-layer interaction.** The cross-covariance argument (Corollary 3.8) establishes that the inner core $|p| \leq 2\ell$ dominates the empirical cross-covariance by a fixed fraction $\beta \geq 0.9$, with the boundary layer ($2\ell < |p| \leq 2\alpha\ell$) contributing $\|\Sigma_{\text{cross}}^{\text{bdry}}\| \leq C\ell^2$ controlled by Gaussian-weight smallness. While this ensures $\sigma_{\min}(\Sigma_{\text{cross}}) \geq c_2\ell^2$ for some dimensional $c_2 > 0$, the explicit dependence of c_2 on α and dimension is not optimised; a more careful analysis could potentially improve the residual bound from $O(\varepsilon_\tau \alpha \ell)$ to $O(\varepsilon_\tau \ell)$, removing the α dependence.

7.3 The convergence rate is hypothesis-dependent

A reader familiar with concentration-of-measure proofs may have noticed that several seemingly arbitrary choices in the construction of Definition 2.3 are calibrated to yield the polynomial convergence rate stated in Theorem 6.1. We acknowledge this openly: the specific rate is not an intrinsic property of the Hauptvermutung, but rather a feature of how strongly we choose to formalise “approximately Poisson density.”

Remark 7.1 (Family of faithful embedding definitions). The proof of Theorem 6.1 works essentially unchanged for any F2 of the form

$$\delta_D = K_d (\log \rho V_M)^\sigma \cdot (\rho \text{Vol}_g(D))^{-\beta}, \quad (39)$$

for $\sigma, \beta \in (0, 1/2]$. Varying β corresponds to assuming the embedding’s count fluctuations decay at some rate β in volume:

- $\beta = 1/2$ (our choice): matches Poisson statistics exactly. Yields the polynomial rate in Theorem 6.1.
- $\beta < 1/2$: weaker concentration assumption (count fluctuations decay slower than Poisson). Yields a slower convergence rate but is satisfied by a strictly larger class of embeddings.
- $\beta > 1/2$: stronger-than-Poisson concentration. A faster convergence rate would result, but only if one posits an embedding more regular than a Poisson sprinkling (e.g., a determinantal or quasi-periodic point process). The upper bound $\beta \leq 1/2$ matches the natural fluctuation floor of Poisson processes (Lemma 2.6).

In particular, a global constant tolerance ($\delta = \text{constant}$, corresponding formally to $\beta = 0$) gives a logarithmically slow but *still valid* Hauptvermutung statement. The qualitative content of the theorem – “ $\varepsilon \rightarrow 0$ as $\rho\lambda^d \rightarrow \infty$ ” – is preserved across the entire family, with only the rate of vanishing differing.

Remark 7.2 (Caution about phenomenology of the rate). This rate-hypothesis dependence has a substantive methodological implication: phenomenological inferences that depend specifically on the polynomial rate $\rho^{-2/(5d)}$ (or any other specific exponent) should be viewed with caution. The rate is not measurable from a single causal set; it would only be testable across a family of causal sets of varying density, and even then, only relative to a chosen formalisation of “faithful embedding.”

What *is* robust across all reasonable formalisations is the qualitative statement: the continuum-manifold description of spacetime emerges from the discrete causal structure in the regime where the discreteness scale $\rho^{-1/d}$ is much smaller than the curvature scale λ . Quantities derived from this qualitative content (e.g., that a manifold description *exists* and is unique in the limit) are physically meaningful. Quantities that depend on the specific polynomial exponent of convergence (e.g., a specific power-law for “spacetime foam fluctuations”) are artifacts of the formalisation and should not be treated as predictions of the underlying physical theory without further justification.

The same point applies to interpretations of the smoothing scale ℓ as a Wilsonian EFT cutoff (§7.6): the cutoff scale is a parameter of the proof, not a prediction of the underlying theory. Different proof strategies will balance discretisation against curvature differently and produce different optimal ℓ . The existence of *some* intermediate scale at which the continuum description becomes valid is the robust content; the precise scale is not.

7.4 Lower bounds and optimality of the rate

How close is our upper bound to the optimal rate? This question has both mathematical and physical significance: the optimal rate represents the irreducible uncertainty in classical geometry induced by the underlying discrete structure.

The Poisson Cramér–Rao lower bound

Any reconstruction $\hat{g}$ of a Lorentzian metric g from a Poisson sprinkling at density ρ has an information-theoretic lower bound on its error. The Fisher information about g contained in $N = \rho V$ Poisson samples in a volume- V region is of order N (each sample contributes $O(1)$ independent information). By the Cramér–Rao bound, any unbiased estimator of any finite-dimensional functional of g has variance $\Omega(1/N)$, giving error

$$\varepsilon_{\text{CR}} \gtrsim \frac{1}{\sqrt{\rho \ell^d}} \quad (40)$$

at scale ℓ . Optimising against the curvature error ℓ^2/λ^2 (which sets the largest scale at which the flat-space estimator applies) gives $\ell_* = \rho^{-1/(d+4)} \lambda^{4/(d+4)}$, with

$$\varepsilon_{\text{CR}} \gtrsim \rho^{-2/(d+4)} \lambda^{-d/(d+4)}. \quad (41)$$

For $d = 4$, this is $\rho^{-1/4} \lambda^{-2/3}$. *No algorithm can recover the metric to better than this accuracy from Poisson-sprinkled data.*

Gap to our upper bound

Comparing with (37), our upper bound is $\rho^{-2/(5d)}\lambda^{-2/5} \log^{3/2}(\rho V_{\max})$, which for $d = 4$ is $\rho^{-1/10}\lambda^{-2/5}$. The exponents differ:

$$\frac{2}{5d} < \frac{2}{d+4}, \quad \text{e.g. } \frac{1}{10} < \frac{1}{4} \text{ for } d = 4.$$

Our rate is therefore *not* the optimal information-theoretic rate. The gap arises because we extract information from the longest-chain length (the Bollobás–Brightwell rate $1/(2d)$), which is one specific combinatorial functional of the causal set, not the maximum-likelihood estimator that would saturate the Cramér–Rao bound.

Conjecture 7.3 (Optimal Hauptvermutung rate). *The true optimal rate in Theorem 6.1 is the Cramér–Rao bound (41). Equivalently, there exists a (likely non-explicit) estimator $\hat{\Phi}$ of the diffeomorphism using all combinatorial data of C such that*

$$\mathbb{E}\|\hat{\Phi}^*g_2 - g_1\|_{g_1} \leq C_d \rho^{-2/(d+4)} \lambda^{-d/(d+4)}.$$

Quantum corrections to classical geometry

If the causal set is the fundamental object and the Lorentzian manifold is an emergent classical limit, then the Cramér–Rao bound is the *intrinsic uncertainty* in classical geometry induced by the underlying discreteness. Any classical observable $\mathcal{O}[g]$ that depends on the metric carries a corresponding uncertainty

$$\Delta\mathcal{O} \gtrsim \|\nabla_g \mathcal{O}\| \cdot \varepsilon_{\text{CR}} \sim \rho^{-2/(d+4)} \|\nabla_g \mathcal{O}\|. \quad (42)$$

For example: the volume of a region of size L has uncertainty $\Delta V/V \gtrsim \rho^{-2/(d+4)}(L/L_*)^?$ where L_* is the relevant scale; geodesic lengths, curvature invariants, and other geometric quantities each have analogous uncertainties. These can be interpreted as “quantum corrections” to classical geometry, even though the underlying theory is purely statistical (Poisson sprinkling) rather than quantum. Conversely, in any genuine quantum theory of causal sets, the Poisson statistics are already present at the kinematical level, and Cramér–Rao gives the unavoidable classical limit on metric reconstruction independent of any quantum dynamics.

This is a concrete realisation of the slogan “the discreteness scale induces an irreducible uncertainty in the continuum metric”: the uncertainty is precisely $\varepsilon_{\text{CR}} \sim \rho^{-2/(d+4)} \lambda^{-d/(d+4)}$, quantitatively determined by the sprinkling density and the spacetime curvature.

Towards saturating the bound

Proving Conjecture 7.3 would require an estimator using more of the combinatorial structure of C than just longest chains. Two natural candidates:

- (i) *Maximum likelihood*: treat the causal order and counts as Poisson observations of an unknown metric and maximise the likelihood. This is non-explicit and likely intractable but achieves the Cramér–Rao bound by construction.
- (ii) *Higher-order chain statistics*: the number of k -element chains in an interval is a Poisson-distributed random variable with mean depending polynomially on ρ and the metric. Combining estimators across k should recover more information than longest chain alone.

We leave the development of such estimators to future work.

7.5 Relation to prior work

Malament [12] proves exact uniqueness from the full causal structure: a continuous, time-orientation-preserving bijection of $d \geq 3$ -dimensional globally hyperbolic spacetimes preserving causal order is necessarily a smooth conformal isometry. Together with volume information this would determine the full metric, and Malament’s result is the conceptual origin of the CST slogan “order plus number gives geometry.”

Müller [13] examined what happens when one tries to formalise this slogan for finite causal sets. He considered two versions:

(i) A direct *finite-set* version, where a finite poset is faithfully embedded into two spacetimes by maps preserving only the causal order. Müller showed this version is *false*: he constructs adversarial examples of finite causal sets admitting causal-order-preserving embeddings into geometrically distinct manifolds. The underlying issue is that without volume information, finite causal data is too sparse to pin down the manifold; an adversary can rearrange the embedded points in different spacetimes while preserving the order.

(ii) A *countable-set* version with stronger hypotheses: the embedded points form a dense subset of the manifold, and the embedding is required to satisfy a “Planck-scale uniform density” condition that is deterministic rather than Poisson. Under these stronger hypotheses, Müller proves an exact isometry result. However, the deterministic uniform-density condition does not correspond to Poisson sprinkling, and the countable limit removes the finite-density structure that is essential to CST.

Our result fills the gap Müller’s work leaves open: we work with finite causal sets, with volume information (F2) encoding the statistical Poisson structure (rather than a deterministic Planck-scale condition), and obtain quantitative approximate uniqueness with an explicit high-density limit. The volume condition is precisely the “number” half of the CST slogan, and our error bounds make precise what “approximate” means as $\rho \rightarrow \infty$.

The Brightwell–Gregory [6] longest chain estimator, originally introduced as a dimension estimator, here plays a structural role: it provides the combinatorial invariant that bridges the two embeddings.

The moving Karcher mean construction may be of independent interest. It simultaneously serves as: (a) a smooth, global interpolation of noisy point correspondences on Riemannian manifolds, defined by a single intrinsic minimisation problem; and (b) a statistical low-pass filter that suppresses high-frequency noise in the derivative through the Delta-method analysis of weighted ratios.

7.6 Physical interpretation: the smoothing scale as an EFT cutoff

The smoothing scale ℓ admits a suggestive interpretation in the language of effective field theory (EFT), though as discussed in §7.3 this interpretation should be understood as a heuristic correspondence rather than a physical prediction. The Hauptvermutung asserts that the continuum metric g is recoverable from the discrete causal set, but only above a resolution scale ℓ . Below this scale, the manifold description breaks down and only the combinatorial structure of C remains.

The EFT cutoff (heuristic correspondence). The scale hierarchy $\rho^{-1/d} \ll \ell \ll L \ll \lambda$ mirrors the structure of a Wilsonian effective field theory: $\rho^{-1/d}$ is the *UV cutoff* (the Planck scale, where discreteness becomes manifest); ℓ is the *smoothing scale* (the scale above which

the continuum description is valid); L is the *mesoscopic scale* (the scale at which curvature effects enter); λ is the *IR curvature scale*. The diffeomorphism Φ is the “coarse-grained” map that captures the continuum physics; the Poisson noise at scales $< \ell$ is the short-distance physics that is integrated out. We emphasise that the specific optimal value of ℓ in our proof ($\ell_* = \rho^{-1/(5d)}\lambda^{4/5}$) is determined by which concentration inequalities we used, not by an underlying physical principle; alternative proof strategies could yield different optimal scales while reaching the same qualitative conclusion.

Spacetime foam: suggestive analogy, not prediction. The stochastic error in the conformal rigidity (Proposition 4.2) is analogous in form to Wheeler’s “spacetime foam” picture: the term $O((\rho\ell^d)^{-1/2})$ in the angular error represents Poisson fluctuations of the discrete causal structure around the smooth continuum, and these fluctuations are *self-averaging* when coarse-grained to scale ℓ . This is consistent with the broad qualitative idea that spacetime geometry emerges from a fluctuating discrete substructure.

However, we caution against reading specific physical predictions out of the precise rate. As discussed in Remark 7.2, the polynomial exponent $1/2$ in the conformal-rigidity stochastic term reflects our choice of Poisson concentration assumptions (Definition 2.3), not an intrinsic property of an underlying physical theory. A genuine quantum gravity theory predicting the structure of spacetime foam would need to specify both the discrete substructure (which causal sets formalise) *and* its quantum-mechanical fluctuations (which causal set theory currently does through Classical Sequential Growth and related dynamics, not through the static sprinkling considered here).

The robust content of the result is therefore: *there exists some intermediate “coarse-graining” scale at which the discrete substructure averages to a well-defined continuum manifold, and the manifold is unique up to errors that vanish in the high-density limit.* The existence and uniqueness statements are physically meaningful; the specific rate is a feature of the proof.

7.7 Universality beyond causal set theory

Although our result is formulated in the language of causal set theory, the underlying logic is more general. The Hauptvermutung uses only two inputs from the discrete structure: *causal order* (a partial order) and *counting* (the number of elements in regions). Any physical theory formulated on a Lorentzian manifold with a fundamental discreteness scale gives rise to a causal set by sprinkling, and the Hauptvermutung implies that this causal set essentially determines the manifold.

This applies to any framework where spacetime has a causal structure and a discrete substructure—not only causal set theory, but also lattice gauge theory (on a causal lattice), spin foam models, and other approaches to quantum gravity. The result is a statement about the *information content* of causal structure plus counting: together, they contain enough information to reconstruct the full Lorentzian geometry up to errors controlled by the density.

The scale separation condition $\lambda \gg \rho^{-1/d}$ defines the regime of validity of the manifold description. When this condition is violated—near singularities, at the Planck scale, or in regions of extreme curvature—the Hauptvermutung error diverges, signalling that the continuum approximation breaks down. The causal set remains well-defined; only its interpretation as a manifold fails. This gives a precise, quantitative criterion for where the manifold description of physics is valid and where it must be replaced by a fundamentally discrete one.

A The cutoff-Gaussian correction

The cutoff weight $w(p) := e^{-|p|^2/\ell^2} \chi(|p|/\alpha\ell)$ used in Construction 3.2 is identical to the pure Gaussian $g(p) := e^{-|p|^2/\ell^2}$ on the inner support $|p| \leq \alpha\ell$, and zero on $|p| \geq 2\alpha\ell$. The two differ only in the annulus $\alpha\ell \leq |p| \leq 2\alpha\ell$, where $0 \leq w(p) \leq g(p) \leq e^{-\alpha^2}$. This appendix documents the resulting corrections, which appear as $O(e^{-\alpha^2})$ terms throughout the proof.

Lemma A.1 (Cutoff-Gaussian deviation). *Let $\{p_k\}$ be a Poisson process at density ρ in d -dimensional Euclidean space (or, more generally, in a Riemannian manifold of curvature $\leq 1/\lambda^2$ with $\ell \ll \lambda$). For $\alpha \geq 1$, define*

$$\begin{aligned} W_{\text{Gauss}} &:= \sum_k g(p_k - x), \\ W_{\text{cut}} &:= \sum_k w(p_k - x), \\ W_{\text{ann}} &:= \sum_{\alpha\ell \leq |p_k - x| \leq 2\alpha\ell} g(p_k - x). \end{aligned}$$

Then:

- (i) **Pointwise bound:** $0 \leq W_{\text{Gauss}} - W_{\text{cut}} \leq W_{\text{ann}} + W_{\text{out}}$, where $W_{\text{out}} := \sum_{|p_k - x| \geq 2\alpha\ell} g(p_k - x) \leq e^{-4\alpha^2} \cdot (\text{count})$.
- (ii) **Expected annulus weight:**

$$\frac{\mathbb{E}[W_{\text{ann}}]}{\mathbb{E}[W_{\text{Gauss}}]} = \frac{\Gamma(d/2, \alpha^2) - \Gamma(d/2, 4\alpha^2)}{\Gamma(d/2)} \sim \alpha^{d-2} e^{-\alpha^2} \quad \text{as } \alpha \rightarrow \infty,$$

where $\Gamma(s, x)$ denotes the upper incomplete gamma function.

- (iii) **Numerical bound:** For $\alpha = 4$ and $d = 4$, the expected annulus fraction is $\approx 16 \cdot e^{-16} \approx 1.8 \times 10^{-6}$, with W_{out} adding negligibly (its expectation scales as $e^{-4\alpha^2}$).

Proof. Part (i). Direct from the definitions: in the inner ball $w = g$, in the annulus $0 \leq w \leq g$, and beyond $w = 0$ while $g > 0$.

Part (ii). In normal coordinates centred at x , the volume element is $dV_h = (1 + O(r^2/\lambda^2)) d^d r$. Computing in spherical coordinates with $\Omega_{d-1} = 2\pi^{d/2}/\Gamma(d/2)$:

$$\begin{aligned} \mathbb{E}[W_{\text{Gauss}}] &= \rho (\ell\sqrt{\pi})^d (1 + O(\ell^2/\lambda^2)), \\ \mathbb{E}[W_{\text{ann}}] &= \rho \Omega_{d-1} \int_{\alpha\ell}^{2\alpha\ell} r^{d-1} e^{-r^2/\ell^2} dr (1 + O(\ell^2/\lambda^2)). \end{aligned}$$

Substituting $u = r^2/\ell^2$ converts the radial integral to $(\ell^d/2)[\Gamma(d/2, \alpha^2) - \Gamma(d/2, 4\alpha^2)]$. The asymptotic $\Gamma(s, x) \sim x^{s-1}e^{-x}$ for large x then gives the stated rate.

Part (iii). Substitute $\alpha = 4$, $d = 4$: $\Gamma(2, 16) \approx 17 \cdot e^{-16}$ (using $\Gamma(2, x) = (x+1)e^{-x}$). Subtracting $\Gamma(2, 64)$ which is negligible gives the stated value. $\square$

Remark A.2 (Propagation in the proof). Throughout the proof we invoke “ $O(e^{-\alpha^2})$ ” corrections in three places:

- *Source covariance* (Lemma 3.15): $\bar{\Sigma} = (\ell^2/2)I + O(e^{-\alpha^2}\ell^2)$.

- *Derivative of weight* (Proposition 3.5(iv)): bump function contribution is $O(e^{-\alpha^2}/\ell)$ pointwise.
- *Integration by parts* (Corollary 3.8): the bump-function term contributes $O(e^{-\alpha^2})$ relative to the leading term.

In each case, the correction comes from the fact that the cutoff weight differs from the pure Gaussian only in the annulus $\alpha\ell \leq |p| \leq 2\alpha\ell$, where the underlying Gaussian factor $e^{-|p|^2/\ell^2}$ is itself $\leq e^{-\alpha^2}$.

For $\alpha \geq 4$, the $O(e^{-\alpha^2}) \leq 10^{-6}$ corrections are utterly dominated by every other error term in the paper (the statistical $O((\rho\ell^d)^{-1/2})$, the curvature $O(\ell^2/\lambda^2)$, the BG longest-chain rate $O((\rho V)^{-1/(2d)})$, etc.). We therefore absorb these corrections into the leading constants without further comment in the main text.

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